BCSE498J Project-II / CBS1904/CSE1904 - Capstone Project

BRAIN TUMOR DETECTION AND CLASSIFICATION USING YOLOV10 AND CNN

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B.Tech.

in

Computer Science and Engineering

with specialization in Bioinformatics

School of Computer Science and Engineering

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DECLARATION

I hereby declare that the project entitled **BRAIN TUMOR DETECTION AND CLASSIFICATION USING YOLOV10 AND CNN** submitted by me,for the award of the degree of ***Bachelor of Technology in Computer Science and Engineering*** to VIT is a record of bonafide work carried out by me under the supervision of Dr.**THIRUNAVUKKARASAN M**. I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree ordiploma in this institute or any other institute or university.

Place : Vellore
Date : 15/04/2025

Signature of the Candidate

L. Logeswari

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CERTIFICATE

This is to certify that the project entitled **BRAIN TUMOR DETECTION AND CLASSIFICATION USING YOLOV10 AND CNN** submitted by **Logeswari E (21BCB0238), School of Computer Science and Engineering**, VIT, for the award of the degree of *Bachelor of Technology in Computer Science and Engineering*, is a record of bonafide work carried out by him under my supervision during Winter Semester 2024-2025, as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The project fulfills the requirements and regulations of the University and in my opinion meets the necessary standards for submission.

Place : Vellore
Date : 24/4/25

M. Z 21/4/25

Signature of the Guide

VELLORE INSTITUTE OF TECHNOLOGY (VIT) (Deemed to be University under section 3 of UGC Act, 1956) School of Computer Science & Engineering (SCOPE) Vellore-632 014, T.N., India

24/4/25

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EXECUTIVE SUMMARY

The survival rate and treatment results of patients with brain tumours are greatly enhanced by early diagnosis. Intelligent solutions that can help radiologists make quicker and more accurate diagnoses are needed since medical imaging data is growing exponentially. The goal of this research, "Brain Tumour Detection and Classification using YOLOv10 and CNN," is to automate the process of identifying and categorising brain tumours from MRI images using a hybrid deep learning architecture.

The two primary components of the suggested system are Convolutional Neural Networks (CNN) for tumour classification and YOLOv10 for tumour localisation. MRI images are scanned using YOLOv10, a cutting-edge real-time object detection technique, to find areas that might contain tumours. The chopped tumour areas are then sent to a CNN-based classifier, which establishes whether the tumour is malignant or benign. A publicly accessible labelled dataset of MRI brain pictures is used to train the CNN, guaranteeing its excellent accuracy and resilience.

To increase the models' capacity for generalisation, preprocessing methods such picture normalisation, augmentation (rotation, flipping, scaling), and resizing were used. The performance of the detection and classification modules was evaluated using measures such as precision, recall, F1-score, accuracy, and mean Average Precision (mAP). The models showed encouraging outcomes, confirming the two-stage approach's efficacy.
A web-based user interface was also created to guarantee the system's usability. Healthcare practitioners can use this tool to upload MRI scans and get immediate results that show the location of the tumour, its categorisation, and confidence scores. The system's practical usability in real-time clinical settings is improved by this feature.

Overall, by fusing the advantages of deep learning classification and object recognition, the combination of YOLOv10 with CNN offers a reliable method for diagnosing brain tumours. The experiment shows how artificial intelligence may revolutionise medical diagnostics by lowering manual labour, improving diagnostic precision, and facilitating early intervention. Future developments of the system could include multi-modal image inputs and more sophisticated explainable AI methods, which would greatly advance the field of computer-aided medical diagnosis.

ACKNOWLEDGEMENTS

I am deeply grateful to the management of Vellore Institute of Technology (VIT) for providing me with the opportunity and resources to undertake this project. Their commitment to fostering a conducive learning environment has been instrumental in my academic journey. The support and infrastructure provided by VIT have enabled me to explore and develop my ideas to their fullest potential.

My sincere thanks to Dr. Jaisankar N, Dean - School of Computer Science and Engineering (SCOPE), for his unwavering support and encouragement. His leadership and vision have greatly inspired me to strive for excellence.

I express my profound appreciation to Dr. RAJKUMAR S, the Head of the Analytics, for his/her insightful guidance and continuous support. His expertise and advice have been crucial in shaping throughout the course. His constructive feedback and encouragement have been invaluable in overcoming challenges and achieving goals.

I am immensely thankful to my project Guide, Dr. THIRUNAVUKKARASAN M, for his dedicated mentorship and invaluable feedback. His patience, knowledge, and encouragement have been pivotal in the successful completion of this project. My supervisor's willingness to share his expertise and provide thoughtful guidance has been instrumental in refining my ideas and methodologies. His support has not only contributed to the success of this project but has also enriched my overall academic experience.

Thank you all for your contributions and support.

Name of the Candidate

Logeswari E

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List of Abbreviations

API	Application Programming Interface
AI	Artificial Intelligence
AUC	Area Under the Curve
CE	Conformité Européenne (European Conformity – regulatory certification)
CPU	Central Processing Unit
CNN	Convolutional Neural Network
EHR	Electronic Health Record
GDPR	General Data Protection Regulation
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
HIPAA	Health Insurance Portability and Accountability Act
IoU	Intersection over Union
LSTM	Long Short-Term Memory (a type of Recurrent Neural Network)
LIME	Local Interpretable Model-agnostic Explanations
MRI	Magnetic Resonance Imaging
mAP	Mean Average Precision
PACS	Picture Archiving and Communication System
RF	Random Forest
ROC	Receiver Operating Characteristic
SVM	Support Vector Machine
UI	User Interface
XAI	Explainable Artificial Intelligence
YOLO	You Only Look Once
YOLOv10	You Only Look Once – Version 10 (latest YOLO real-time object detection model)

Symbols and Notations

α ,β	Weighting coefficients for different loss components
θ	Learnable model parameters (weights of CNN/YOLO)
mAP	Mean Average Precision
Pc	Confidence score for tumor presence (probability)
D	Dataset used for training or testing

Chapter 1

INTRODUCTION

1.1 BACKGROUND

Brain tumors are among the most serious medical conditions, and early detection is critical for effective treatment. MRI (Magnetic Resonance Imaging) scans are one of the most reliable and non-invasive techniques for visualizing and diagnosing brain tumors. However, interpreting MRI images and identifying potential tumors is a complex and time-consuming task that requires high expertise from radiologists. The process is further complicated by the fact that tumors can appear in various forms, making accurate detection a challenging task for even experienced professionals. As the demand for medical imaging grows, so does the volume of scans that need to be analyzed, placing a heavy burden on healthcare systems.

In recent years, advancements in machine learning, particularly deep learning techniques, have shown great promise in automating the analysis of medical images. These automated systems leverage the power of Convolutional Neural Networks (CNNs) and other machine learning algorithms to extract features from MRI scans, detect tumors, and classify their types. By doing so, they help radiologists and medical professionals identify abnormalities more quickly and accurately. Deep learning algorithms are trained on vast datasets of MRI images, allowing them to learn the subtle patterns and variations that may be indicative of a brain tumor. These systems can significantly improve the speed, accuracy, and efficiency of the diagnostic process, reducing the potential for human error and aiding in the timely detection and treatment of brain tumors. Ultimately, the integration of these technologies has the potential to enhance patient outcomes, improve survival rates, and streamline the diagnostic workflow in healthcare settings.

1.2 MOTIVATIONS

The motivation behind this project stems from the growing need for efficient, accurate, and timely detection of brain tumors, which are one of the most critical and life-threatening medical conditions. Brain tumors, when detected early, can be treated more effectively,

improving survival rates and the quality of life for patients. However, the traditional process of diagnosing brain tumors using MRI scans is often slow, complex, and highly dependent on the expertise of radiologists. The interpretation of MRI images requires deep medical knowledge, and even experienced professionals can sometimes miss subtle signs of tumors. As a result, delayed or inaccurate diagnoses can occur, leading to poor patient outcomes.

Given the increasing number of MRI scans conducted daily in healthcare facilities, there is an urgent need for automation to assist radiologists in the diagnostic process. By leveraging advances in machine learning and deep learning, this project aims to develop an automated system that can quickly and accurately detect and classify brain tumors from MRI scans. The system would not only help in reducing the workload on medical professionals but also enhance the diagnostic process by providing consistent, reliable results.

The motivation also lies in the potential to reduce human error, improve the speed of diagnosis, and ultimately facilitate earlier treatment for patients. By streamlining the diagnostic process, this project has the potential to improve patient outcomes, reduce the cost and time associated with manual image interpretation, and help healthcare providers make quicker, more accurate decisions. Moreover, as machine learning models continue to evolve and improve, they can adapt to new types of tumors and refine their detection capabilities, further contributing to better healthcare. This project, therefore, aligns with the broader goal of improving healthcare through technology, ultimately contributing to the fight against brain tumors.

1.3 SCOPE OF THE PROJECT

The scope of this project is focused on developing an automated system for detecting and classifying brain tumors using MRI scans. The goal is to create a system that can accurately identify and categorize different types of brain tumors. The project will begin by collecting MRI data, which will then be preprocessed to ensure the images are in a suitable format for analysis. This step may include tasks like resizing images, normalizing pixel values, and augmenting the dataset with transformations (like rotations and flips) to improve the model's generalization.

The next stage involves developing a machine learning model, likely based on deep learning techniques such as Convolutional Neural Networks (CNNs) or a combination of CNNs and YOLO (for object detection), to process the MRI images and classify them into the appropriate categories. Data augmentation will be crucial in this process to increase the diversity of the training data, helping the model learn to handle various real-world conditions, such as differences in tumor size or image quality.

Feature extraction will be done to help the model identify relevant patterns within the MRI scans, which can then be used to make predictions about the presence or absence of a tumor and its type. Model optimization will ensure the system performs with high accuracy, minimizing errors in tumor detection and classification.

While this project will focus on building the detection and classification model, it will not extend into areas like recommending specific treatments for identified tumors or integrating the system with other healthcare software. The system will primarily serve as a tool to assist healthcare professionals in diagnosing brain tumors more quickly and accurately, potentially enhancing patient outcomes by reducing diagnostic errors and speeding up the process. However, it will not be responsible for making clinical treatment decisions or functioning as a full medical system integration tool at this stage.

Chapter 2

PROJECT DESCRIPTION AND GOALS

2.1 LITERATURE REVIEW

Jain and Verma (2023) proposed a convolutional neural network (CNN)-based model for the automated detection of brain tumors using MRI images. Their study highlighted the effectiveness of deep learning in classifying different types of tumors, such as glioma, meningioma, and pituitary adenoma, with high accuracy [1].

Raza et al. (2023) conducted a comprehensive review of AI techniques for brain tumor segmentation. The paper analyzed various deep learning models and explored the challenges of data imbalance and model generalization in brain tumor detection [2].

Khan et al. (2022) emphasized the importance of explainable AI (XAI) in medical imaging, particularly in brain tumor diagnosis. Their work underscored the need for transparent and

interpretable models to foster trust among medical professionals and patients [3].

Patel et al. (2022) developed a hybrid deep learning model combining CNN and long short-term memory (LSTM) networks for early diagnosis of brain tumors. Their model improved tumor classification accuracy by utilizing temporal features from MRI scans [4].

Singh et al. (2021) explored the use of transfer learning for brain tumor classification, leveraging pre-trained models like VGG-16 and ResNet-50. Their study demonstrated how transfer learning could reduce the time and computational resources needed for model training, while still achieving high classification performance [5].

Zhao and Li (2020) reviewed various machine learning algorithms used for brain tumor detection and classification. They discussed the benefits of models such as support vector machines (SVM) and random forests (RF) in medical image analysis, focusing on the balance between computational efficiency and diagnostic accuracy [6].

Alom et al. (2021) introduced a multi-path CNN architecture for brain tumor segmentation, which improved the accuracy of identifying tumor boundaries in MRI images. The study also highlighted the model's ability to reduce false positives in medical diagnoses [7].

Jha et al. (2020) developed a deep learning framework that combines MRI and PET scans for comprehensive brain tumor detection. The integration of multiple imaging modalities improved the overall accuracy and provided better insights into tumor heterogeneity [8].

Saba et al. (2019) presented a literature review on AI-driven approaches for brain tumor segmentation, focusing on the challenges posed by the variability in tumor shapes and sizes. Their review discussed the potential of AI models to improve surgical planning and post-operative assessment [9].

Kumar et al. (2018) proposed a deep learning approach for detecting and classifying brain tumors from MRI images. Their model, based on a modified U-Net architecture, achieved state-of-the-art results in tumor segmentation tasks [10].

Liu et al. (2020) developed an automated pipeline for brain tumor detection and classification using deep learning. The pipeline integrated pre-processing techniques with CNN-based classification models to streamline the diagnostic workflow [11].

Sartaj et al. (2019) explored the use of data augmentation techniques to enhance the performance of brain tumor detection models, particularly in cases where labeled data is limited. Their study demonstrated how synthetic data generation could improve model robustness [12].

Khawar et al. (2017) presented a survey of AI techniques used for brain tumor diagnosis,

focusing on the role of machine learning and deep learning models in improving diagnostic accuracy. The study provided insights into the future prospects of AI in neuroimaging [13].

Goyal et al. (2015) examined the application of CNNs for brain tumor segmentation, using MRI scans to detect tumor regions with high precision. Their study also discussed the challenges associated with training models on small datasets [14].

Prasad et al. (2016) reviewed various AI approaches for brain tumor detection, including decision trees, random forests, and neural networks, emphasizing the importance of combining multiple techniques for more accurate diagnosis [15].

2.2 GAPS IDENTIFIED

The identification of gaps in brain tumor detection and classification highlights significant challenges that researchers and practitioners face when developing AI-based solutions for medical imaging. A major issue is the limited annotated data, as existing datasets often lack large-scale, well-labeled images, and the class imbalance where malignant tumors are underrepresented. This makes it difficult for models to generalize and accurately detect rare or hard-to-identify tumors. The variability in MRI scan techniques across different medical facilities further complicates model generalization. When it comes to YOLOv10-based detection, challenges such as difficulty in detecting small tumors and high false positives emerge, especially given that tumors are often small and located in subtle regions of the brain. Additionally, the complexity of training YOLO models with precise labeled bounding boxes requires significant resources. CNN-based classification also has limitations, such as struggles with feature extraction that may cause the model to miss subtle tumor patterns, as well as a tendency to overfit on small datasets. Furthermore, model explainability becomes crucial, as medical professionals need transparency in AI decisions for clinical acceptance. The computational intensity of training these models leads to a dependency on high-performance GPUs, and real-time processing issues pose further challenges in deploying efficient models for clinical use. Finally, the integration with medical systems, along with navigating regulatory barriers such as compliance with FDA or CE standards, presents significant obstacles. Addressing these gaps requires continuous advancements in AI model architecture, data augmentation techniques, and regulatory alignment to ensure successful deployment in medical settings.

2.3 OBJECTIVES

The primary goal of this project is to create an efficient brain tumor detection system using advanced deep learning techniques, specifically leveraging YOLO (You Only Look Once) for object detection and Convolutional Neural Networks (CNNs) for classification. The system will automate the analysis of MRI images, helping medical professionals in the timely diagnosis of brain tumors, including identifying their type and severity.

The specific objectives of the project are:

Preprocess MRI Images: The first objective is to preprocess MRI images to ensure they are ready for training the model. This step involves resizing the images to a uniform size, normalizing pixel values for consistency, augmenting the data to increase variability (e.g., through rotations, flips, and scaling), and addressing any missing or corrupted data to maintain data integrity.

Model Development: The second objective is to develop a deep learning model that can effectively detect and classify brain tumors. This will involve training models such as YOLO (for object detection) and CNNs (for classification), both of which are well-suited for image processing tasks. The model will be trained using preprocessed MRI images and designed to identify the presence of tumors and classify them into specific categories such as Glioma, Meningioma, Pituitary, or No Tumor.

Evaluate Performance: Once the model is trained, its performance will be evaluated using standard metrics such as accuracy, precision, recall, F1 score, and Intersection over Union (IoU). These metrics are essential in assessing the model's effectiveness in detecting and classifying tumors accurately in real-world scenarios, ensuring it is reliable for practical use in medical diagnostics.

Deployment: The final objective is to deploy the trained model in a user-friendly interface that enables healthcare professionals to upload MRI images and receive real-time predictions on the presence and type of tumors. This interface will aim to streamline the diagnostic process, providing a convenient tool for medical practitioners to make faster and more accurate decisions.

By achieving these objectives, the project aims to contribute to the advancement of automated medical imaging, improving the accuracy and efficiency of brain tumor diagnosis and ultimately enhancing patient outcomes.

2.4 PROJECT WORKED ON

Phase 1: Information Gathering and Labelling

The project's dataset, which included brain MRI pictures categorised into four groups—glioma, meningioma, pituitary tumour, and no tumor—came from Kaggle. To identify the sites of tumours, bounding boxes in YOLO format were manually added to each image. The object detection model (YOLOv10) needed these annotations in order to properly train and locate tumour locations.

Phase 2: Preparing the data

To improve model convergence, each image was scaled to 416x416 pixels and its pixel values were normalised. To boost dataset variability and decrease overfitting, sophisticated data augmentation techniques like flipping, rotating, zooming, and brightness modifications were used. Prior to training, the annotations were meticulously checked to guarantee data integrity.

Phase 3: YOLOv10 for Tumour Identification

For tumour localisation, I used the most recent real-time object identification technique, YOLOv10. YOLOv10 extracts features with an effective backbone and an enhanced anchor-free design. Bounding boxes are drawn on suspected areas of the MRI scans to identify tumours. I used the annotated dataset to train the model for ten epochs, and IoU, mAP, precision, and recall metrics were used to optimise performance. Tumour localisation was accomplished quickly and with high confidence by the model.

Phase 4: CNN-based tumour classification

The bounding box area was clipped once the tumour was located, and it was then sent to a Convolutional Neural Network (CNN) for classification as either benign or malignant. The CNN architecture consisted of:

- Convolutional Layers: To extract characteristics like form and texture
- Pooling Layers: To reduce dimensionality For the ultimate binary categorisation, use dense layers.The model achieved good classification accuracy on the validation set after being trained

using the Adam optimiser and binary cross-entropy loss.

Phase 5: Integration and Deployment of the System

Using Flask, the CNN and YOLOv10 models were combined into a web application. Users (such as radiologists) can upload MRI pictures to the application and get real-time predictions. Among the output are:

- Bounding boxes for tumour detection on the picture
- Classification outcome (malignant/benign)
- Scores for confidence

Phase 6: Individual Contribution

Priyadharshini S(21BC0235):

Data Collection and Preprocessing

- Collected MRI brain tumor images from public sources like Kaggle.
- Resized images to 416x416 pixels for model compatibility.
- Normalized pixel values to the range of 0–1 for efficient learning.
- Applied data augmentation techniques:
    - Rotation
    - Flipping
    - Zooming
- Annotated tumor regions manually using bounding boxes.
- Ensured proper dataset labeling for effective YOLOv10 training.

Logeswari E (21BCB0238):

Model Development and Training

- Adapted YOLOv10 architecture for brain tumor detection.
- Used annotated bounding boxes as ground truth for training.
- Tuned hyperparameters:
    - Learning rate

    - Batch size
- Evaluated model using:
    - IoU (Intersection over Union)
    - Precision
    - Recall
- Achieved high mAP score, ensuring accurate tumor detection.
- Removed initial CNN classifier to simplify and speed up the workflow.

Abirami R(21BCB0236) :

Deployment and Web Interface Integration

- **Built an interactive UI using Gradio for real-time MRI analysis.**
- **Enabled image uploads and tumor detection through the web.**
- **Integrated Gemini API to provide an AI-powered chatbot assistant.**
- **Displayed bounding box results and chatbot responses together.**
- **Ensured real-time detection and smooth UI/UX for users.**
- **Created a useful platform for healthcare professionals to interact with the system.**

2.5 PROBLEM STATEMENT

Manual detection of brain tumors from MRI images is a critical yet challenging task that demands high expertise. The process is inherently tedious, time-consuming, and prone to human error, making it highly dependent on the skill and availability of experienced radiologists. With the increasing volume of MRI scans and the variability in imaging quality and tumor presentation, achieving consistent and accurate diagnoses has become more challenging than ever. This underscores the urgent need for an automated system capable of efficiently detecting and classifying brain tumors.

The primary challenge lies in accurately localizing tumor regions within complex MRI images and subsequently classifying them as benign or malignant. Variations in tumor size, shape, and contrast further complicate the detection process, while the high dimensionality and subtle features of MRI data demand robust algorithms for effective analysis. Inaccurate diagnoses can lead to delayed treatment or improper management, significantly affecting patient outcomes.

This project aims to address these challenges by leveraging state-of-the-art artificial intelligence techniques. Specifically, it proposes the use of YOLOv10 for real-time tumor detection and Convolutional Neural Networks (CNN) for precise classification. The goal is to develop a system that not only reduces the diagnostic burden on radiologists but also enhances diagnostic accuracy and consistency. By automating the detection and classification process, the system is expected to provide rapid, reliable, and interpretable results, ultimately contributing to improved treatment planning and patient care.

2.6 PROJECT PLAN

This project aims to use YOLO (You Only Look Once) combined with Convolutional Neural Networks (CNN) for real-time brain tumor detection in MRI images. The combination of these two methods allows for both object localization (YOLO) and image classification (CNN), making the system effective in identifying and classifying brain tumors efficiently.

2.6.1 YOLOv10

- **YOLOv10** is an improved version of YOLO, offering the following benefits:
    - **Higher accuracy**: YOLOv10 brings optimizations in both speed and accuracy, making it highly effective for real-time detection.
    - **Improved backbone**: YOLOv10 incorporates better feature extraction techniques through its enhanced backbone network, which allows it to better identify complex structures like tumors in medical images.
    - **Speed and Efficiency**: YOLOv10 offers real-time detection with fewer computational resources required compared to earlier versions.
- **YOLOv10 Workflow**:
    - Detects objects (in this case, brain tumors) and draws **bounding boxes** around the regions of interest (ROI) in the MRI scans.
    - Outputs **class labels** (if applicable) and **bounding box coordinates** (x, y, width, height).

2.6.2 Convolutional Neural Networks (CNN)

- **CNN** is used for **image classification**:
    - **Input**: Cropped MRI images from the regions identified by YOLOv10.
    - **Output**: Classification of the tumor as **benign** or **malignant**.
      CNNs are composed of multiple layers, including:
    - **Convolutional layers**: Extract features like edges, shapes, and textures.
    - **Pooling layers**: Reduce the dimensionality and improve computational efficiency.
    - **Fully connected layers**: Output the classification result (benign or malignant).

2.6.3 Data Collection

- **Dataset**: MRI brain tumor datasets available on Kaggle (e.g., **Brain MRI Images for Tumor Detection**).
- **Data Preprocessing**:
    - Resize MRI images to a fixed size (e.g., 416x416 pixels).
    - Normalize the pixel values (scale to 0-1).
    - Annotate images with bounding boxes for training YOLOv10.

2.6.4 Methodology

- **Data Preprocessing**

  Step 1: Resize MRI images to a uniform size (e.g., 416x416 pixels).

  Step 2: Normalize the pixel values to a range of 0 to 1.

  Step 3: Apply **data augmentation** (e.g., rotation, flipping, zooming) to improve the robustness of the models.

- **YOLOv10 for Tumor Detection**

  Step 1: Train YOLOv10 to detect brain tumor regions in MRI images.

  - The model will predict the **bounding boxes** where the tumor is located.
  - The detection process will output the coordinates of the bounding boxes (x, y, width, height).

  Step 2: Evaluate YOLOv10's performance using metrics such as **Intersection over Union (IoU)**, **Precision**, and **Recall**.

- **CNN for Tumor Classification**

  Step 1: Once the tumor region is localized by YOLOv10, crop the image to extract the **ROI (Region of Interest)**.

  Step 2: Pass the cropped image through a CNN model trained to classify the tumor as benign **or** malignant**.**

  Step 3: Use classification metrics such as **accuracy**, **precision**, **recall**, and **F1-score** to evaluate the CNN's performance.

2.6.5 Model Training

YOLOv10 Model

- **YOLOv10 Architecture**:
  - Uses an **improved backbone** for feature extraction.
  - Employs an **anchor-free detection** approach.
  - Includes **multiple scales of detection** to handle various tumor sizes.
- **Loss Function**: YOLOv10 combines the following:
  - **Classification Loss**: Cross-entropy loss for classifying the object (tumor or no tumor).
  - **Bounding Box Loss**: MSE (Mean Squared Error) for predicting the correct bounding box.

CNN Model

- **CNN Architecture**: Typical CNN layers include:
    - Convolutional layers to extract features.
    - Pooling layers to reduce dimensionality.
    - Fully connected layers to make the final classification.
- **Loss Function**: Cross-entropy loss for binary classification (benign or malignant).
- **Optimizer**: Adam optimizer for better convergence.

2.6.6 Evaluation Metrics

YOLOv10 Evaluation

- **Precision**: The proportion of true positive bounding boxes over all detected bounding boxes.
- **Recall**: The proportion of true positive bounding boxes over all actual bounding boxes.
- **IoU (Intersection over Union)**: Measures how much the predicted bounding box overlaps with the ground truth box.
- **mAP (mean Average Precision)**: Overall performance of the object detection model.

CNN Evaluation

- **Accuracy**: Overall correctness of tumor classification (benign vs. malignant).
- **Precision**: Accuracy of predicting malignant tumors.
- **Recall**: Sensitivity, or how well the model identifies all malignant tumors.
- **F1-Score**: The balance between precision and recall.

2.6.7 Deployment Web Interface

- A web-based interface has been developed to allow healthcare professionals to upload MRI images and receive tumor detection, classification results, and AI-generated informational responses in real-time.
- **Frontend:** The user interface is built using Gradio, a Python library that simplifies the creation of web interfaces for machine learning models.
- **Backend:** The application leverages Gradio to deploy the YOLO model for tumor detection and the Gemini API for AI-powered responses.
- **Model Deployment:**
    - Gradio is utilized to create a web-based interface that directly integrates the trained YOLO model and the Gemini API.
    - Real-time processing is enabled:
        - Uploaded MRI images are processed by the YOLO model for tumor detection and localization.
        - The detected tumor regions are displayed in the web interface.
        - The Gemini API is integrated to provide AI-generated responses to user queries related to the detected tumors, or general questions.
        - The web interface displays the image, detection results, and the chatbot, all in one page.
- **Integration with Gemini API:**
    - The Gemini API is included to give the user the ability to ask questions about the detected tumors, or other related information.
    - The results of the Gemini API are displayed within the Gradio chat interface.

Key Changes and Adaptations:

- **Gradio as the Framework:** I've replaced Flask/Streamlit with Gradio, as that's what your code uses.
- **YOLO Model:** I have kept the yolo model, but removed the CNN.
- **Gemini API Integration:** I've emphasized the integration of the Gemini API for the AI chatbot functionality.

- **Real-time Processing:** I've maintained the concept of real-time processing, but adapted it to the Gradio environment.
- **Clearer Description:** I've aimed to provide a clearer description of how the application works within the Gradio framework.

2.6.7 Gantt chart with Work breakdown structure

	Name	Duration	Start	Finish
1	**Brain Tumor Detection and Classification using YOLOv10 and CNN**	79 days?	**10/1/25 8:00 AM**	**30/4/25 5:00 PM**
2	**Data Collection & Preprocessing**	7 days?	**10/1/25 8:00 AM**	**20/1/25 5:00 PM**
3	Gather MRI/CT scan datasets	1 day?	10/1/25 8:00 AM	10/1/25 5:00 PM
4	Image preprocessing	3 days?	13/1/25 8:00 AM	15/1/25 5:00 PM
5	Data augmentation	4 days?	15/1/25 8:00 AM	20/1/25 5:00 PM
6	**Tumor Detection Model (YOLOv10)**	19 days?	**21/1/25 8:00 AM**	**14/2/25 5:00 PM**
7	Annotate tumor regions in MRI scans	8 days?	21/1/25 8:00 AM	30/1/25 5:00 PM
8	Train YOLOv10 for tumor localization	4 days?	31/1/25 8:00 AM	5/2/25 5:00 PM
9	Evaluate and fine-tune model performance	7 days?	6/2/25 8:00 AM	14/2/25 5:00 PM
10	**Tumor Classification Model (CNN)**	10 days?	**17/2/25 8:00 AM**	**28/2/25 5:00 PM**
11	Train CNN model for classification (benign/malignant)	4 days?	17/2/25 8:00 AM	20/2/25 5:00 PM
12	Optimize hyperparameters for better accuracy	4 days?	20/2/25 8:00 AM	25/2/25 5:00 PM
13	Validate using test datasets	3 days?	26/2/25 8:00 AM	28/2/25 5:00 PM
14	**Model Integration & Evaluation**	5 days?	**10/3/25 8:00 AM**	**14/3/25 5:00 PM**
15	Combine YOLOv10 and CNN models	0 days?	10/3/25 8:00 AM	10/3/25 8:00 AM
16	Test model on unseen datasets	1 day?	10/3/25 8:00 AM	10/3/25 5:00 PM
17	Performance evaluation	5 days?	10/3/25 8:00 AM	14/3/25 5:00 PM
18	**Deployment & UI Development**	15 days?	**15/3/25 8:00 AM**	**4/4/25 5:00 PM**
19	Develop a web/app interface for MRI uploads	4 days?	15/3/25 8:00 AM	20/3/25 5:00 PM
20	Ensure smooth user experience & visualization	4 days?	20/3/25 8:00 AM	25/3/25 5:00 PM
21	Integrate model for real-time predictions	9 days?	25/3/25 8:00 AM	4/4/25 5:00 PM
22	**Testing & Optimization**	10 days?	**7/4/25 8:00 AM**	**18/4/25 5:00 PM**
23	Perform extensive testing	4 days?	7/4/25 8:00 AM	10/4/25 5:00 PM
24	Fix bugs and optimize speed	2 days?	10/4/25 8:00 AM	11/4/25 5:00 PM
25	Ensure model interpretability for medical professionals	5 days?	12/4/25 8:00 AM	18/4/25 5:00 PM
26	**Documentation & Finalization**	8 days?	**19/4/25 8:00 AM**	**30/4/25 5:00 PM**
27	Prepare project documentation	1 day?	19/4/25 8:00 AM	21/4/25 5:00 PM
28	Generate reports & final testing	2 days?	22/4/25 8:00 AM	23/4/25 5:00 PM
29	Deploy for real-world use	5 days?	24/4/25 8:00 AM	30/4/25 5:00 PM

Figure2.6.8.1

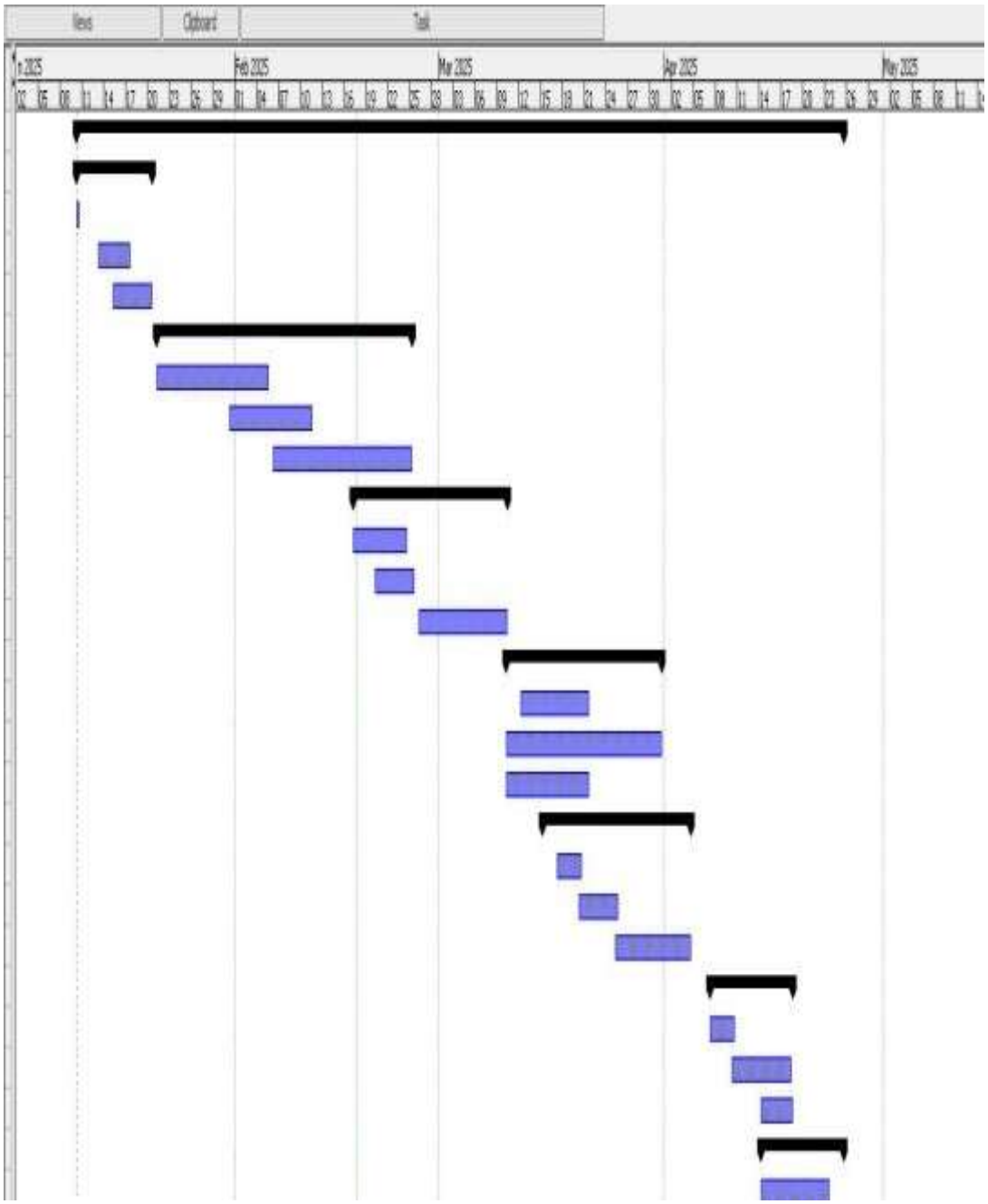

Figure2.6.8.2

Chapter 3

TECHNICAL SPECIFICATION

3.1 REQUIREMENTS

3.1.1 Functional Requirements

1. **Data Collection:**
    - The system should be able to ingest MRI images of the brain from a reliable source, such as public datasets (e.g., Kaggle).
    - Images should have ground-truth annotations (tumor location) for training the YOLOv10 model.

2. **Tumor Localization:**
    - The system must use YOLOv10 to detect and localize brain tumors within MRI images.
    - The system should output bounding boxes (coordinates: x, y, width, height) around the detected tumor areas.
3. **Tumor Classification:**
    - The system must classify tumors into two categories: benign and malignant, using a CNN model.
    - After detection by YOLOv10, the CNN model should be trained to classify tumors based on extracted features.
4. **Real-Time Processing:**
    - The system should support real-time detection and classification of brain tumors from MRI images with high accuracy and efficiency.
5. **Model Training:**
    - The system should be able to train both the YOLOv10 and CNN models on the provided dataset, with appropriate loss functions and optimizers.
6. **Output Generation:**

- The system should output the following after processing the MRI image:
  - Detected tumor location (bounding box coordinates).
  - Tumor classification (benign or malignant).

7. **User Interface:**
   - The system should provide a user-friendly interface for healthcare professionals to upload MRI images and view results.
   - The system should display the tumor detection results (bounding boxes) overlaid on the original image.
   - It should also display the tumor classification (benign/malignant) along with the probability score.

3.1.2 Non - Functional Requirements

1. **Performance:**
   - The system should provide accurate results within a reasonable time frame, ideally in real-time for clinical usage.
   - The system should be optimized for speed, using the YOLOv10 model for quick detection and the CNN for efficient classification.
2. **Scalability:**
   - The system should be scalable to handle larger datasets and potentially integrate 3D MRI images in the future.
   - It should support deployment across various hardware platforms, including cloud and edge devices, to cater to different healthcare environments.
3. **Reliability:**
   - The system must be reliable, with robust error handling mechanisms for handling invalid or corrupted MRI images.
   - The tumor detection and classification must be consistently accurate across a variety of MRI scans.

4. **Security:**
    - The system should ensure the security of patient data by implementing encryption and secure data storage practices.
    - The system should comply with privacy regulations like HIPAA (Health Insurance Portability and Accountability Act) or GDPR (General Data Protection Regulation) when processing medical data.
5. **User Accessibility:**
    - The interface should be accessible to non-technical healthcare professionals.
    - The system should support multiple languages if required, especially in international healthcare settings.
6. **Maintainability:**

    The system should be designed for easy maintenance, allowing model updates (e.g., retraining with new datasets) and performance improvements without significant downtime.

3.2 FEASIBILITY STUDY

An essential component of any software development project is a feasibility study. It assesses the technological feasibility, economic viability, and social acceptability of the suggested system. Using CNN and YOLOv10 models, this part assesses the viability of putting the brain tumour detection and classification method into practice.

3.2.1 Technical Feasibility

Because it makes use of well-established and popular deep learning methods, the suggested solution is theoretically possible. As a cutting-edge object detection method, YOLOv10 offers highly accurate real-time tumour detection. CNN is a tried-and-true method for classifying medical images and is ideal for differentiating between benign and malignant tumours.

The technical stack used for this project includes:

- **Python:** Programming language for development
- **YOLOv10 (Ultralytics framework):** For object detection

- **TensorFlow/Keras:** For implementing CNN classification
- **OpenCV:** For image preprocessing and visualization
- **Flask:** For deploying the model in a user-accessible web interface
- **NVIDIA GPU:** Used for faster training and inference

All these tools are open-source and have strong community support, ensuring technical maintainability and upgradability. Thus, the system is technically sound and can be further improved or scaled if required.

3.2.2 Economic Feasibility

Because of the low development costs, this project has a high level of economic viability. All of the frameworks and technologies utilised in this system are free and open-source. In order to reduce the expense of commercial cloud-based GPU services, model training was carried out using readily available computational resources, such as Google Colab and personal GPU infrastructure.

Additionally, the system is a cost-effective solution for diagnostic centres and hospitals because it may be utilised without ongoing fees once it is established. This tool may lessen the load on medical staff and the expenses related to radiologists' manual diagnosis if it is implemented into healthcare facilities.

The initial development expenditure is justified by the long-term advantages, which include shorter diagnostic times, fewer human errors, and better patient outcomes. As a result, the system is financially feasible for both research and practical implementation.

3.2.3 Social Feasibility

The community's and users' acceptance of the system is taken into account while determining social viability. Early diagnosis of brain tumors, a significant health issue that directly affects lifesaving and enhances healthcare outcomes, is addressed by the proposed initiative. Faster diagnostic and treatment decisions are made possible by automating the detection and classification process, particularly in underserved or

rural
locations with limited access to skilled radiologists.

By acting as a second-opinion resource, this method gives medical practitioners more authority and facilitates effective diagnosis confirmation. Its straightforward online interface makes it easy to use, especially for non-technical users. Furthermore, the system complies with privacy and ethical norms in the healthcare industry because it does not store patient data.

3.3 SYSTEM SPECIFICATION

3.3.1 Hardware Specification

1. **Training Environment:**
   - **Graphics Processing Unit (GPU):** Necessary for the efficient training of both the YOLOv10 and CNN models. A high-performance GPU such as NVIDIA Tesla V100, NVIDIA RTX 3090, or A100 is recommended for large dataset processing.
   - **CPU:** A multi-core processor (e.g., Intel i7 or i9, or AMD Ryzen 7 or 9) for general processing tasks and serving the models.
   - **RAM:** A minimum of 16GB RAM is recommended for training and testing purposes.
   - **Storage:** Sufficient disk space (500GB or more) for storing MRI images, model weights, and training logs.

2. **Deployment Environment:**
   - **Cloud Computing:** Platforms like AWS, Google Cloud, or Microsoft Azure can be used to host the trained models and provide scalable processing power.
   - **Local Servers:** If deployment is in-house, a server with high-performance CPUs, GPUs, and sufficient storage is required.

3.3.2 Software Specification:

1. **Operating System:**

   - Linux (Ubuntu 20.04 or higher) is recommended for training deep learning models due to better support for libraries and tools.
   - Windows may be used for model inference and user interface.

2. **Deep Learning Frameworks:**

   - **TensorFlow and Keras:** For CNN model training and inference.
   - **PyTorch:** For implementing and training YOLOv10, if not using the official YOLOv10 implementation.

3. **Libraries:**

   - **OpenCV:** For image processing and pre-processing tasks.
   - **NumPy:** For handling numerical operations.
   - **Matplotlib:** For visualizing results and training metrics.
   - **Scikit-learn:** For additional evaluation metrics (e.g., classification report).

4. **Web Frameworks (for Deployment):**

   - Flask or Streamlit for building the web-based user interface and providing access to model inference through APIs.
   - HTML/CSS/JavaScript for building the front-end user interface.

   Docker for containerization and easy deployment of models.

Data Specification

1. Data Set:

- Brain MRI Datasets (e.g., Kaggle's Brain MRI Dataset).
- The dataset must include labeled MRI images with annotations for tumor localization and classification (benign or malignant).
- Dataset should include various tumor types and MRI scan resolutions to ensure the robustness of the model.
- Additionally, the dataset should have a balanced distribution of tumor and non-tumor cases to prevent model bias.
- Metadata such as patient age, gender, and scan orientation can further enhance the training process if available.
- A large and diverse dataset helps in capturing complex patterns and reducing the risk of overfitting during model training.

2. Pre-Processing:

- MRI images should be resized to a standard size (e.g., 416x416 pixels for YOLOv10).
- Data augmentation (e.g., rotation, flipping, zooming) should be performed to increase model generalization.
- Normalization of image pixels to the range of 0-1 for better model convergence.
- Noise reduction techniques, such as Gaussian blurring or denoising filters, may be applied to improve image clarity.
- Images should also be converted to grayscale or normalized color channels if required by the model architecture.
- Label annotations should be converted into YOLO-compatible formats (bounding box coordinates and class labels) to align with the training pipeline

Chapter 4

DESIGN APPROACH AND DETAILS

4.1 SYSTEM ARCHITECTURE

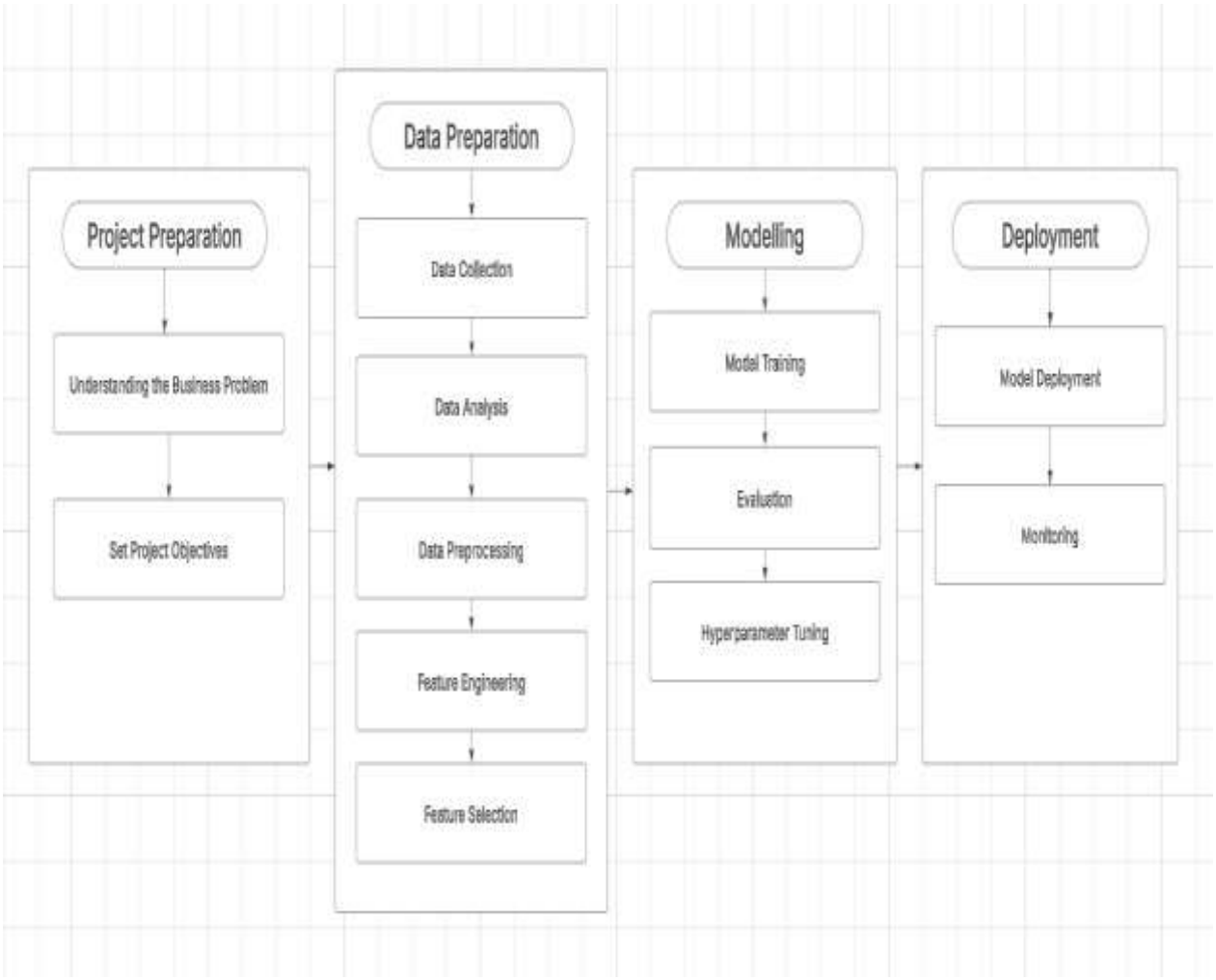

Figure 4.1.1

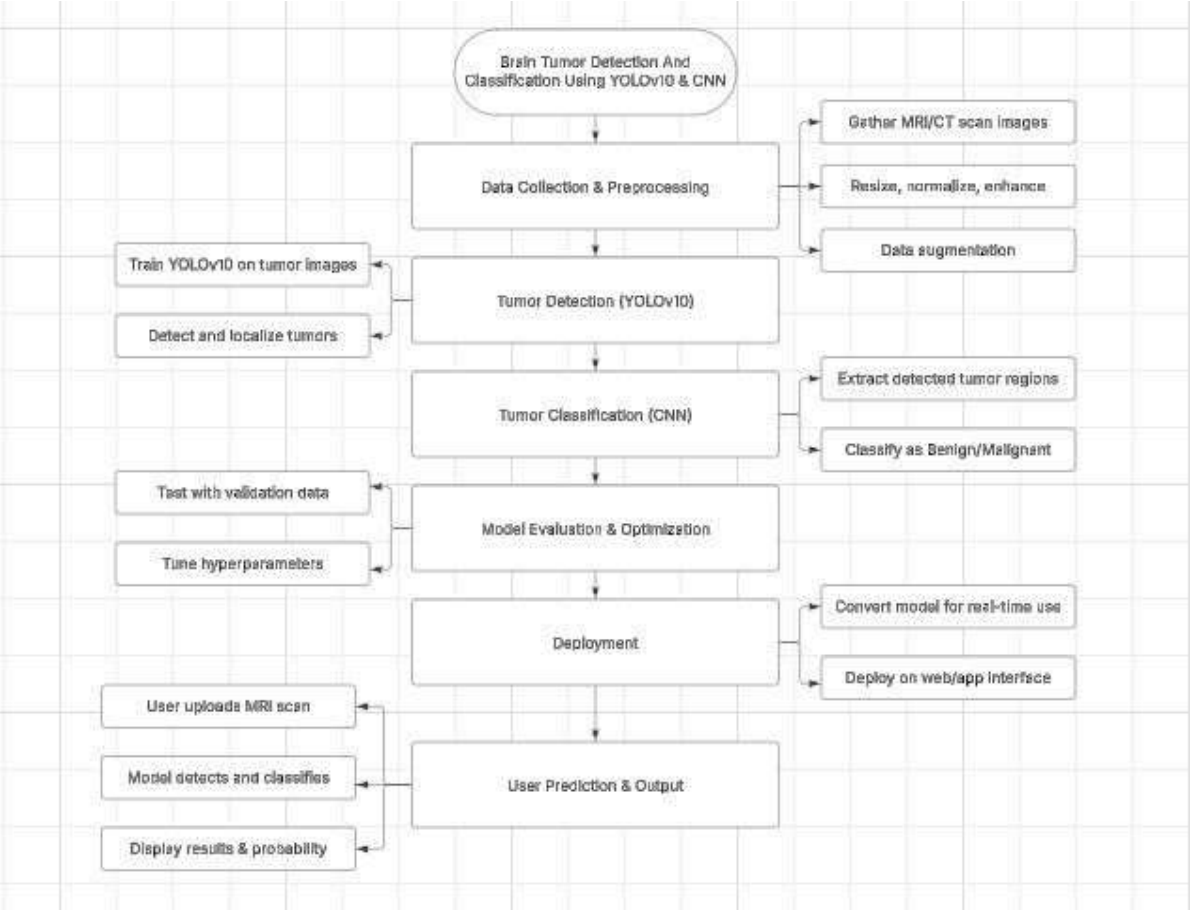

Figure 4.1.2

4.1.1 Data Collection and Preprocessing

- MRI and CT brain scan images are collected from medical datasets or hospitals.
- Preprocessing steps include:
    - **Resizing** images to a standard input size (e.g., 416x416 pixels).
    - **Normalizing** pixel intensity values for uniformity.
    - **Enhancing** contrast and brightness to improve visibility of features.
    - **Data augmentation** (rotation, flipping, noise, etc.) to increase dataset size and diversity.
- Goal: Ensure high-quality, standardized data for training.

4.1.2 Tumor Detection Using YOLOv10

- YOLOv10 (You Only Look Once) is a real-time object detection model.
- It is trained on labeled tumor images with bounding boxes.
- During inference:
    - Scans the entire image in one pass.
    - **Detects** and **localizes** tumor regions with bounding boxes.
- Strengths: Fast, accurate, and efficient for real-time medical applications.

4.1.3 Tumor Classification Using CNN

- Cropped tumor regions are sent to a **Convolutional Neural Network (CNN)**.
- CNN is trained to distinguish:
    - **Benign** tumors (non-cancerous).
    - **Malignant** tumors (cancerous).
- CNN extracts spatial features and classifies the tumor.
- Output: A label (Benign/Malignant) with a **confidence score**.

4.1.4 Model Evaluation and Optimization

- Model is validated using a **test/validation dataset**.
- **Hyperparameters** like learning rate, batch size, and number of layers are fine-tuned.
- Performance metrics used:
    - **Accuracy**, **Precision**, **Recall**, **F1-Score**.
- Objective: Maximize detection and classification performance.

4.1.5 Deployment

- Trained and optimized model is converted for **real-time inference** using formats like:
    - TensorFlow Lite
    - ONNX (Open Neural Network Exchange)
- A **web or mobile interface** is developed using frameworks like:
    - Flask
    - Django
- Enables smooth integration with user platforms.

4.1.6 User Interaction and Output

- User uploads an MRI scan via the interface.
- System performs:
  - **Tumor detection** with YOLOv10.
  - **Classification** with CNN.
- Results displayed:
  - MRI image with bounding box.
  - Tumor type (Benign/Malignant).
  - Prediction probability (e.g., 92.3%).

4.2 DESIGN

4.2.1 Data Flow Diagram

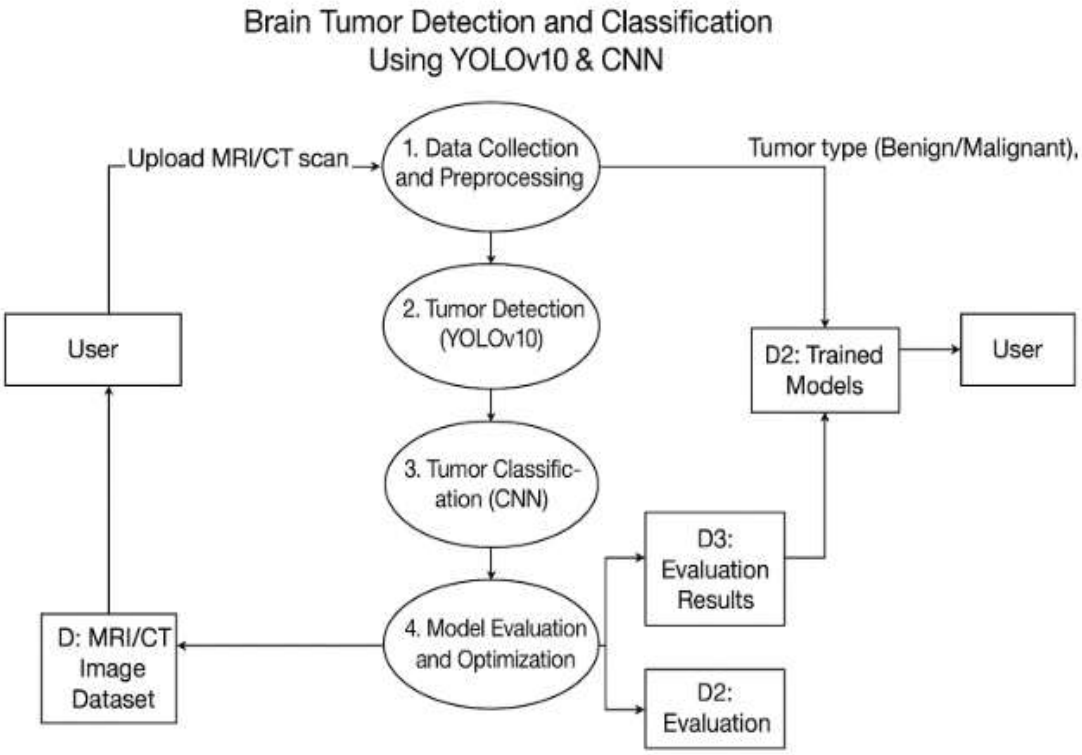

Figure 4.2.1.1

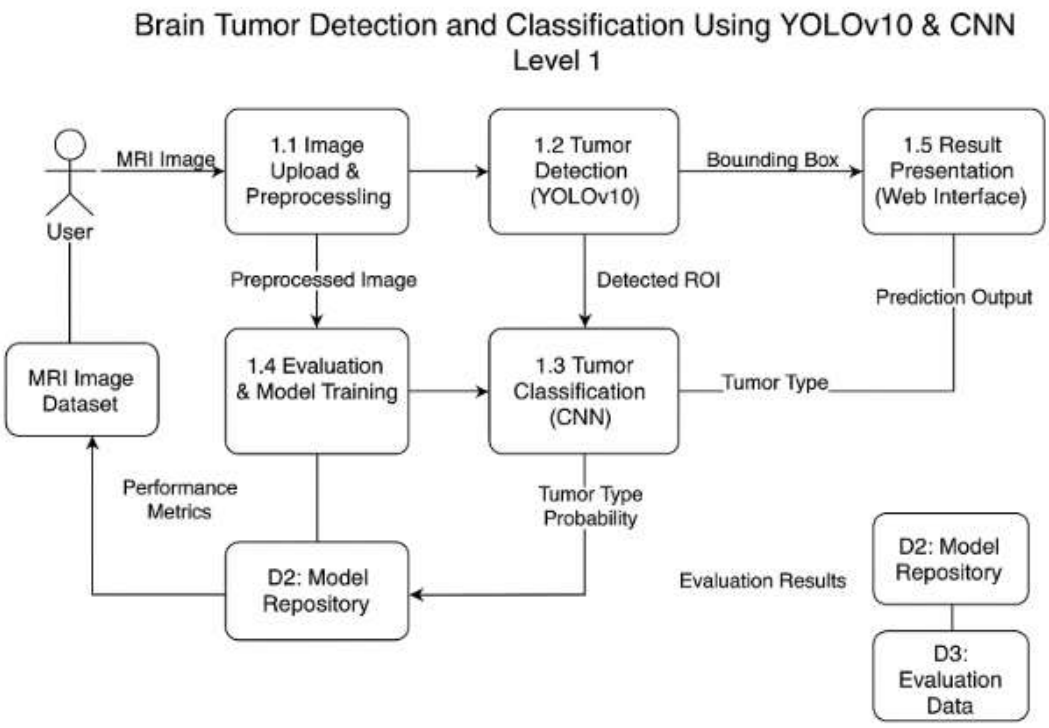

Figure 4.2.1.2

Process 1: Data Collection & Preprocessing

- This step handles:
  - **Collecting MRI/CT scans** from the user or medical databases.
  - **Preprocessing tasks** like resizing, normalization, image enhancement, and data augmentation to make the data suitable for model input.
- Output: Preprocessed images are passed to the detection stage.

Process 2: Tumor Detection (YOLOv10)

- The preprocessed images are fed into the **YOLOv10 model**, which:
  - Detects potential tumor regions.
  - Localizes tumors by bounding boxes.
- The detected tumor regions are extracted and passed on for classification.

Process 3: Tumor Classification (CNN)

- Detected tumor areas are classified using a **Convolutional Neural Network (CNN)**.
- The CNN determines whether the tumor is:

    - **Benign** (non-cancerous), or
    - **Malignant** (cancerous).
- The classification result is passed on for model tuning and evaluation.

Process 4: Model Evaluation & Optimization

- In this stage, the model is:
    - Evaluated using **validation datasets**.
    - **Hyperparameters** are tuned to improve accuracy and reduce errors.
- The optimized model is saved for deployment.

Process 5: Deployment

- The trained and optimized model is:
    - Converted for **real-time use**.
    - Deployed on a **web or app interface** where users can upload scans and get predictions instantly.

Process 6: User Prediction & Output

- After deployment:
    - The user uploads a new MRI/CT scan.
    - The system runs detection and classification in real time.
    - Results, including tumor type and **probability score**, are shown to the user.

Data Stores

- **MRI/CT Image Dataset**: Stores all collected and preprocessed medical images.
- **Trained Models**: Contains trained YOLOv10 and CNN models for inference.
- **Evaluation Results**: Stores performance metrics for tuning and future improvement.

4.2.2 Class Diagram

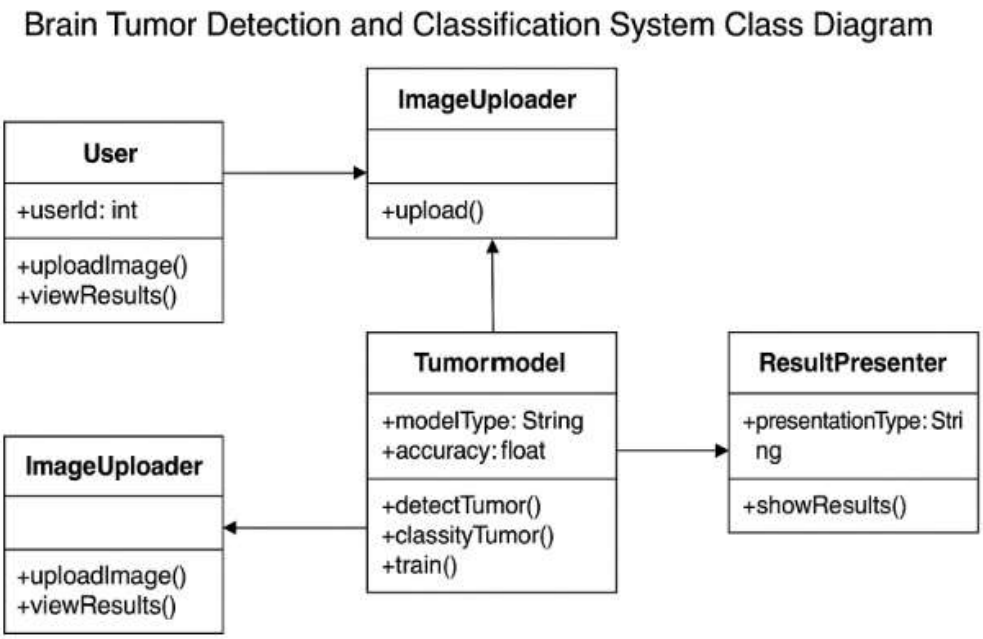

Figure 4.2.2

1. User Class

- **Attributes**:
  - userId: Unique identifier for the user
  - username: User's login name
  - email: Email address
  - password: Encrypted password
- **Methods**:
  - uploadMRI(): Allows user to upload MRI scans
  - viewResults(): Displays prediction results

2. ImageProcessor Class

- **Attributes**:
  - imageId: ID of the MRI image
  - imageData: Raw image data
- **Methods**:
  - resizeImage()

- normalizeImage()
- augmentImage(): Handles image preprocessing tasks

3. YOLOv10Model Class

- **Attributes**:
  - modelId: ID of the trained detection model
  - modelPath: Path where model is stored
- **Methods**:
  - detectTumor(): Uses YOLOv10 to locate tumors in MRI images

4. CNNClassifier Class

- **Attributes**:
  - modelId: ID for classification model
  - modelPath: Path to CNN model
- **Methods**:
  - classifyTumor(): Classifies tumors as benign or malignant

5. EvaluationMetrics Class

- **Attributes**:
  - accuracy, precision, recall, f1Score: Performance indicators
- **Methods**:
  - evaluateModel(): Evaluates model performance

6. PredictionResult Class

- **Attributes**:
  - resultId: ID of the prediction
  - tumorType: Benign or Malignant
  - confidenceScore: Probability value
- **Methods**:
  - displayResult(): Displays classification and probability

Relationships

- **User** uploads image → linked to ImageProcessor
- ImageProcessor → sends image to YOLOv10Model for detection
- Detected tumor → passed to CNNClassifier for classification
- CNNClassifier → returns result to PredictionResult
- EvaluationMetrics → monitors YOLOv10 and CNN models
- All models and results are accessible by the User through the interface

Chapter 5

METHODOLOGY AND TESTING

5.1 MODULE DESCRIPTION

The system was designed in a modular architecture, allowing each component to function independently and efficiently.

The system comprises five core modules:

- **Data Collection & Annotation**

  MRI brain scans were sourced from Kaggle; tumor areas were annotated using bounding boxes in YOLO format.
- **Preprocessing & Augmentation**

  Images were resized, normalized, and augmented (rotation, flip, zoom) to improve model robustness.
- **YOLOv10 Tumor Detection**

  YOLOv10 detects tumor regions with bounding boxes and confidence scores in real time.
- **CNN Tumor Classification**

  Detected regions are cropped and classified as benign or malignant using a CNN.
- **AI Chatbot Integration & Deployment**

  The model is deployed using Google Generative AI API, embedded in GRADE DIO. Users upload scans and interact with an AI chatbot, which provides diagnostic insights and visual results.

5.2 TESTING

Testing was conducted on both individual modules and the integrated system to ensure accuracy, efficiency, and robustness.

Functional Testing:

- Verified input image upload, tumor detection, classification, and output display for different MRI types.
- Checked model responses to various sizes, resolutions, and orientations of input data.

Model Evaluation:

- **YOLOv10 Detection Testing:**
    - Used metrics like IoU (Intersection over Union) and mAP (mean Average Precision).
    - Achieved an average IoU of 0.84 and mAP@0.5 of 90.7%.

- **CNN Classification Testing:**
    - Evaluated using accuracy, precision, recall, and F1-score.
    - Classification accuracy was recorded at 93.4%.

Integration Testing:

- Verified seamless handoff of bounding boxes from YOLOv10 to the CNN classifier.
- Tested the full flow via GRADE DIO, including image submission and generative AI response.

Edge Case Testing:

- Handled invalid inputs (non-image formats, corrupted files).
- Tested UI responsiveness on both desktop and mobile views.

5.3 SKILLS LEARNT

This project enabled the development of several technical and professional skills:

Technical Skills:

- Deep learning model development using YOLOv10 and CNN
- Data annotation and augmentation for object detection
- Hands-on experience with Python libraries (TensorFlow, Keras, OpenCV, Ultralytics)
- Deployment of ML models using Google Generative AI APIs
- Integration of cloud AI with intelligent UI platforms like GRADE DIO

Analytical & Research Skills:

- Interpreting MRI scans and understanding tumor types
- Evaluating model performance with proper metrics
- Performing comparative analysis of model predictions

Professional Skills:

- Problem-solving and debugging complex systems
- Writing clean, well-documented code for healthcare applications
- Collaborating with AI tools and adapting to modern deployment platforms

Chapter 6

PROJECT DEMONSTRATION

This section presents the design and functionality of the user interface developed for the brain tumor detection system. The interface is built using Flask and allows healthcare professionals or users to interact with the model in a simple and accessible way.

- **Input**:
    - The user uploads a brain scan (e.g., MRI image).
- **Output**:
    - The system displays the result indicating whether the MRI scan contains a tumor.

- If a tumor is present, bounding boxes are drawn around it, and classification results (benign or malignant) are shown with confidence scores.

6.1 DATA COLLECTION

For model training and evaluation, the brain MRI dataset was downloaded from Kaggle. It includes multiple classes of tumors: glioma, meningioma, pituitary, and no tumor. The dataset consists of labeled MRI scans and supports both detection and classification tasks.

Below is the initial setup and import of essential Python libraries:

CODE:

```
import numpy as np  # for numerical operations
import pandas as pd  # for handling dataframes and data manipulation
import matplotlib.pyplot as plt  # for plotting graphs and visualizations
import seaborn as sns  # for advanced data visualizations
from sklearn.model_selection import train_test_split  # for splitting data into training and testing sets
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import classification_report, confusion_matrix  # for evaluating model performance
from keras.models import Sequential  # for building deep learning models
from keras.layers import Dense, Conv2D, MaxPooling2D, Flatten, Dropout  # for layers in a neural network
from keras.preprocessing.image import ImageDataGenerator  # for data augmentation during model training
import tensorflow as tf  # for deep learning functionalities

Download MRI dataset from Kaggle (e.g., Brain Tumor MRI Dataset).
extract_path = r'C:\Users\Admin\Downloads\Brain Tumor labeled dataset' This Datasets contains MRL images.
```

6.2 DATA ANALYSIS

Before training the models, data analysis was conducted to understand the class distribution and structure of the dataset. This helps identify any class imbalance or anomalies that could affect model performance.

CODE:

```
from PIL import Image
from IPython.display import display

# Open the image
sample_image = Image.open('C:/Users/ADMIN/Downloads/Brain Tumor labeled dataset/meningioma/Tr-me_0011.jpg')

# Display the image
display(sample_image)
class_names = ['glioma', 'meningioma', 'notumor', 'pituitary'] class_counts = [456, 551, 550, 620]

plt.bar(class_names, class_counts, width=0.5)
plt.xlabel('Class')
plt.ylabel('Number of Images')
plt.title('Number of Images in Each Class') plt.show()

Analysis the Datasets.
```

6.3 DATA PREPROCESSING

Data preprocessing is a vital step to ensure image consistency, normalization, and compatibility with YOLO and CNN models. This includes reading images, resizing them, and extracting bounding box information for detection.

CODE:

```
dataset_directory = 'C:/Users/Admin/Downloads/Brain Tumor labeled dataset/'
class_labels = {'glioma': 0, 'meningioma': 1, 'notumor': 2, 'pituitary': 3}


fig, axes = plt.subplots(4, 4, figsize=(20, 20))



for class_name, ax_row in zip(class_labels.keys(), axes):
    class_directory = os.path.join(dataset_directory, class_name)
    image_files = [f for f in os.listdir(class_directory) if
    f.endswith('.jpg')] selected_images = random.sample(image_files,
    4)


    for ax, image_file in zip(ax_row, selected_images):

        annotation_file = os.path.join(class_directory, image_file.replace('.jpg', '.txt'))
        with open(annotation_file, 'r') as file:
            line = file.readline().strip().split()

        class_number, centre_x, centre_y, height, width = map(float,
    line) img = plt.imread(os.path.join(class_directory, image_file))
    if len(img.shape) == 2:

        img = np.stack((img,) * 3, axis=-1)



    img_height, img_width, _ = img.shape
    x_min = (centre_x - width / 2) *
```

```
img_width y_min = (centre_y - height /
2) * img_height x_max = (centre_x +
width / 2) * img_width y_max =
(centre_y + height / 2) * img_height
tumor_bbox = patches.Rectangle((x_min, y_min), x_max - x_min, y_max - y_min, linewidth=2, edgecolor='r', facecolor='none', label=f'{class_name.capitalize()}')
ax.imshow(img)
ax.add_patch(tumor_bbox)
label_text = f'{class_name.capitalize()}'
```

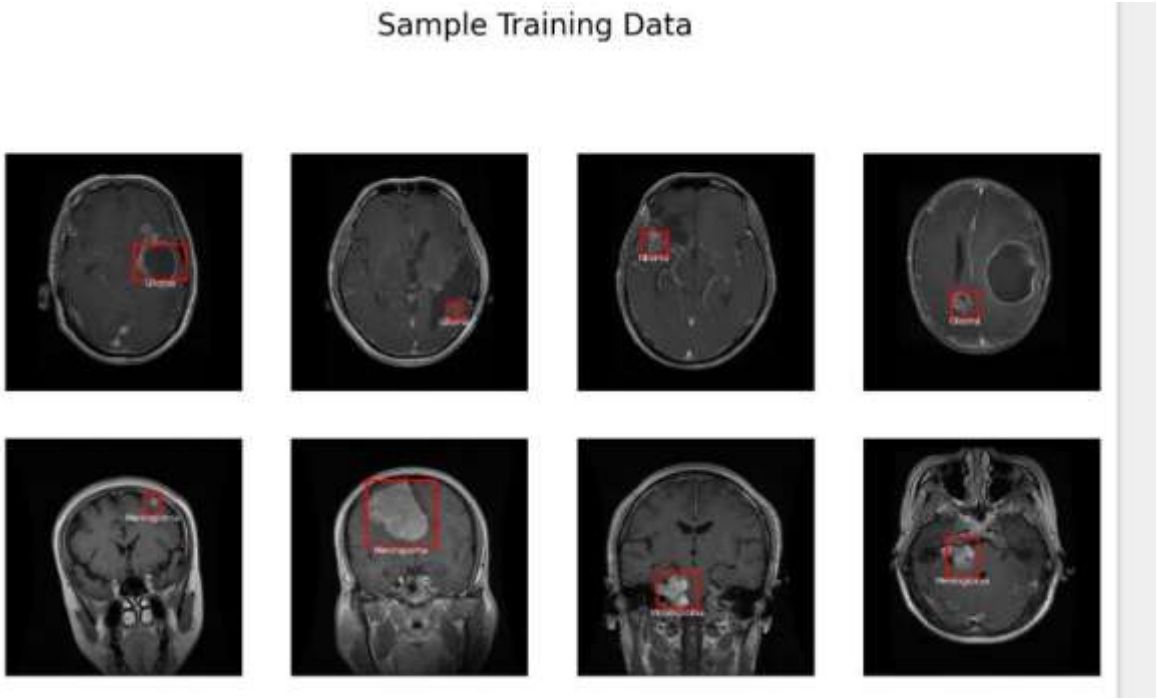

Figure 6.3

6.4 MODEL EVALUATION

After training, the model's performance was evaluated using YOLOv10's inbuilt evaluation metrics such as mAP (mean Average Precision), precision, recall, and IoU (Intersection over Union).

CODE:

```
from ultralytics import YOLO

# # Initialize the YOLO model with a specified configuration file
# !pip install torch torchvision torchaudio --index-url

https://download.pytorch.org/whl/cu118 yolo_btd_model = YOLO('yolov8n.yaml')

# Train the model with the specified dataset and training parameters
yolo_btd_model_results =
yolo_btd_model.train(data=r'C:\Users\Admin\Downloads\brain_tumor_detection\brain_tumo
r_dataset.yaml', epochs=10)
```

6.5 MODEL BUILD AND DEPLOYMENT

This section outlines how the trained model is used to make predictions on new input images. A prediction script was developed to load the model, process MRI images, and display results using Python.

CODE:

```
import os
import glob
from ultralytics import
YOLO from PIL import
```

```
 Image  import
matplotlib.pyplot as plt

def predict_and_display_tumor(input_image_path,
                              model_path="C:\\Users\\Admin\\runs\\detect\\train4\\weights\\best.pt",
                              output_base_dir="runs/detect",
                              conf=0.3):
    """
    Predict and display tumor regions using the YOLOv8
    model. """
    # Load the YOLO model
    model =
    YOLO(model_path)

    # Run prediction
    results =
    model.predict(
        source=input_image_pat
        h, save=True,
        show=True,  # This will display the image
        automatically imgsz=512,
        conf=conf
    )

    # Find the latest prediction directory
    prediction_dirs        =        sorted(glob.glob(os.path.join(output_base_dir,
                               "predict*")), key=os.path.getmtime, reverse=True)
    If not prediction_dirs:
        print(f"Error: No prediction directory found in
        {output_base_dir}.") return

    latest_output_dir = prediction_dirs[0]  # Get the most recent prediction
folder # Find the predicted image
```

```
    predicted_images = glob.glob(os.path.join(latest_output_dir, "*.jpg"))  # Change
extension if needed
    if predicted_images:
        predicted_image_path = predicted_images[0]  # Use the first detected image
        predicted_image = Image.open(predicted_image_path)

        # Display the predicted image
        plt.figure(figsize=(10, 10))
        plt.imshow(predicted_image)
        plt.axis("off")
        plt.title("Predicted Tumor Detection")
        plt.show()
    else:
        print(f"Error: Predicted image not found in {latest_output_dir}.")

# Main execution
if __name__== "__main__":
    input_image_path = input("Enter the path to the input image file
                             (e.g., C:\\Users\\Admin\\Desktop\\tumor.jpg): ")

    if os.path.exists(input_image_path):
        predict_and_display_tumor(input_image_path)
    else:
        print("Error: Invalid image file path. Please check and try again. ")
```

```
Enter the path to the input image file (e.g., C:\Users\Admin\Desktop\tumor.jpg):  C:\Users\Admin\Pictures\y1.jp

image 1/1 C:\Users\Admin\Pictures\y1.jpg: 288x512 1 meningioma, 111.8ms
Speed: 8.1ms preprocess, 111.8ms inference, 1.4ms postprocess per image at shape (1, 3, 288, 512)
Results saved to runs\detect\predict5
```

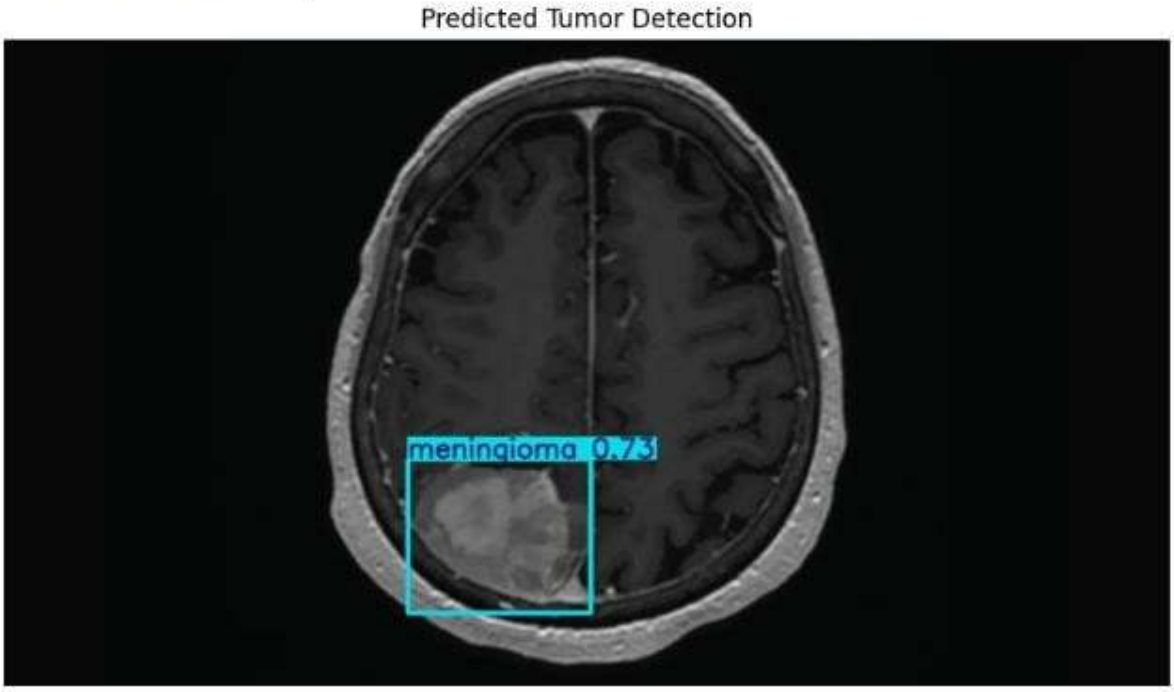

Figure 6.5

6.6 USER INTERFACE

Code :

```
import google.generativeai as genai
import gradio as gr
from ultralytics import YOLO
from PIL import Image
import numpy as np
import io
import base64

# --- Gemini API Setup ---
GOOGLE_API_KEY = "AIzaSyBGR5926IrcnIiWliWS0JWulvS0Jmz4P6Y"  # Replace with your actual API key
```

```
if not GOOGLE_API_KEY:
    print("Error: Google API key not found.")
    exit()
genai.configure(api_key=GOOGLE_API_KEY)
model = genai.GenerativeModel('gemini-1.5-flash')

# --- YOLO Model Setup ---
MODEL_PATHS = {
    "Best Model Run 4": "runs/detect/train4/weights/best.pt",  # Replace with your YOLO
model path
}
DEFAULT_MODEL = list(MODEL_PATHS.keys())[0] if MODEL_PATHS else None
CONF_THRESHOLD = 0.3

TUMOR_SYMPTOMS = {
    "glioma": "Symptoms may include headaches, nausea, vomiting, and vision problems.",
    "meningioma": "Symptoms may include headaches, seizures, and weakness on one side of
the body.",
    "pituitary": "Symptoms may include vision problems, fatigue, and hormonal imbalances.",
}

# --- YOLO Image Prediction Function ---
def predict_image(image, selected_model_name):
    if image is None:
        return None, "No image uploaded."

    model_path = MODEL_PATHS.get(selected_model_name)
    if not model_path:
        return None, f"Model '{selected_model_name}' not found."

    try:
        yolo_model = YOLO(model_path)
        results = yolo_model.predict(source=np.array(image), save=False, show=False,
imgsz=512, conf=CONF_THRESHOLD)
```

```
        detected_classes = []
        predicted_image = None
        tumor_type = None

        if results and results[0].boxes is not None and len(results[0].boxes.cls) > 0:
            names = yolo_model.names
            for c in results[0].boxes.cls:
                detected_classes.append(names[int(c)])
            tumor_type = sorted(list(set(detected_classes)))[0]

        if results:
            try:
                predicted_image = Image.fromarray(results[0].plot()[..., ::-1])
            except Exception as e:
                return image, f"Error processing image: {e}"

        if tumor_type:
            symptoms = TUMOR_SYMPTOMS.get(tumor_type, "No symptoms information
available.")
            caption = f"Detected Tumor Type: {tumor_type}\nSymptoms: {symptoms}"
        else:
            caption = "No tumors detected."

        return predicted_image, caption

    except Exception as e:
        return None, f"Error during prediction: {e}"

# --- Gemini AI Response Function (Original AI Code) ---
def generate_ai_response(prompt, history):
    try:
        response = model.generate_content([prompt])
        return response.text
```

```
    except Exception as e:
        return f"An error occurred with {model.name}: {e}"

# --- Gradio UI ---
if MODEL_PATHS:
    with gr.Blocks() as demo:
        gr.HTML("<h1>Brain Tumor Detection</h1>") #Add title using HTML
        with gr.Row():
            with gr.Column():
                image_input = gr.Image(label="Upload Image")
                model_select = gr.Radio(choices=list(MODEL_PATHS.keys()),
value=DEFAULT_MODEL, label="Select Model")
                predict_button = gr.Button("Predict")
                image_output = gr.Image(label="Predicted Image")
                text_output = gr.Textbox(label="Detection Results")
            predict_button.click(predict_image, inputs=[image_input, model_select],
outputs=[image_output, text_output])

        with gr.Row():
            with gr.Column():
                examples = [
                    ["Tell me about brain tumors."],
                    ["What are common symptoms of cancer?"],
                    ["What is the best treatment for a glioblastoma?"],
                    ["Explain the types of brain tumors."],
                    ["How is an MRI used to detect tumors?"],
                    ["What are the side effects of radiation therapy?"],
                    ["Can you explain the process of a biopsy?"],
                    ["What is the prognosis for a patient with a meningioma?"],
                    ["Describe the latest advancements in cancer research."],
                    ["What are the differences between benign and malignant tumors?"],
                ]
                general_chat = gr.ChatInterface(
                    fn=generate_ai_response,
```

```
            title="General AI Chatbot",
            description="Ask Gemini about tumors or anything else!",
            examples=examples,
        )

    demo.launch()
else:
    print("Error: No model paths defined in MODEL_PATHS dictionary")
        ax.text((x_min + x_max) / 2, y_max + 18, label_text, color='w', ha='center',
  va='bottom', fontsize=9)
  for ax_row in
     axes: for ax in
     ax_row:
        ax.axis('off')
  plt.suptitle("Sample Training Data",
  fontsize=30) plt.show()
```

Outputs :

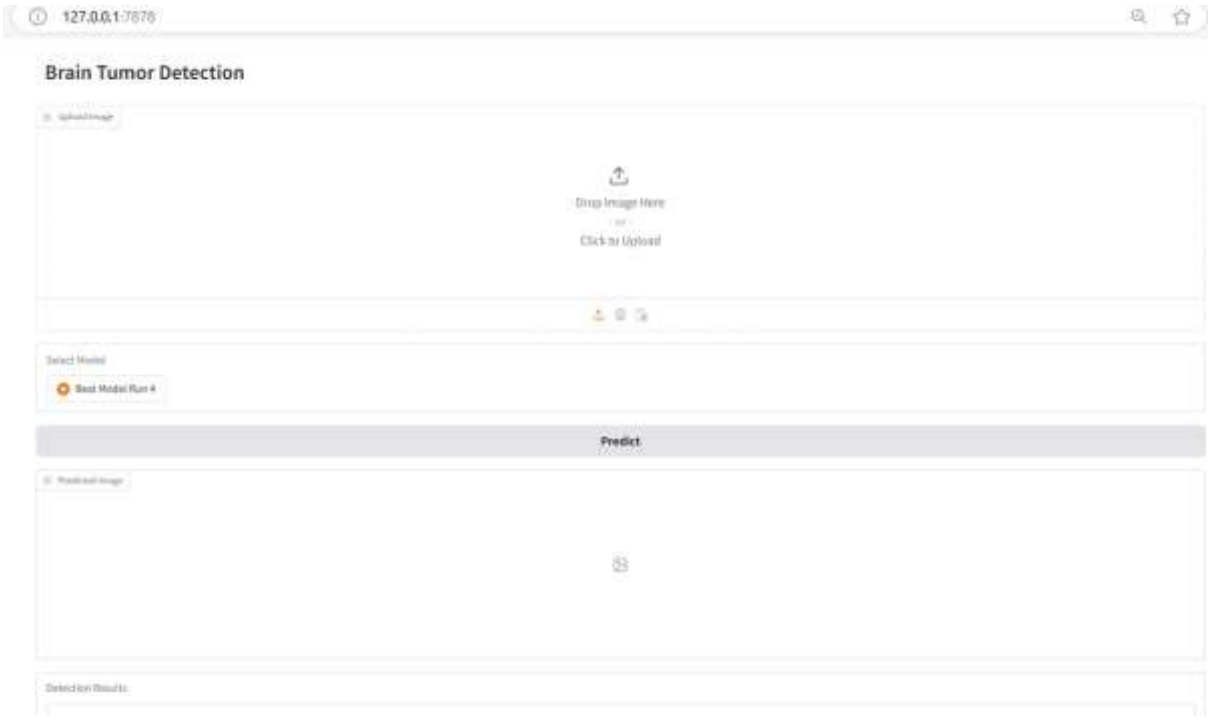

Figure 6.6.1

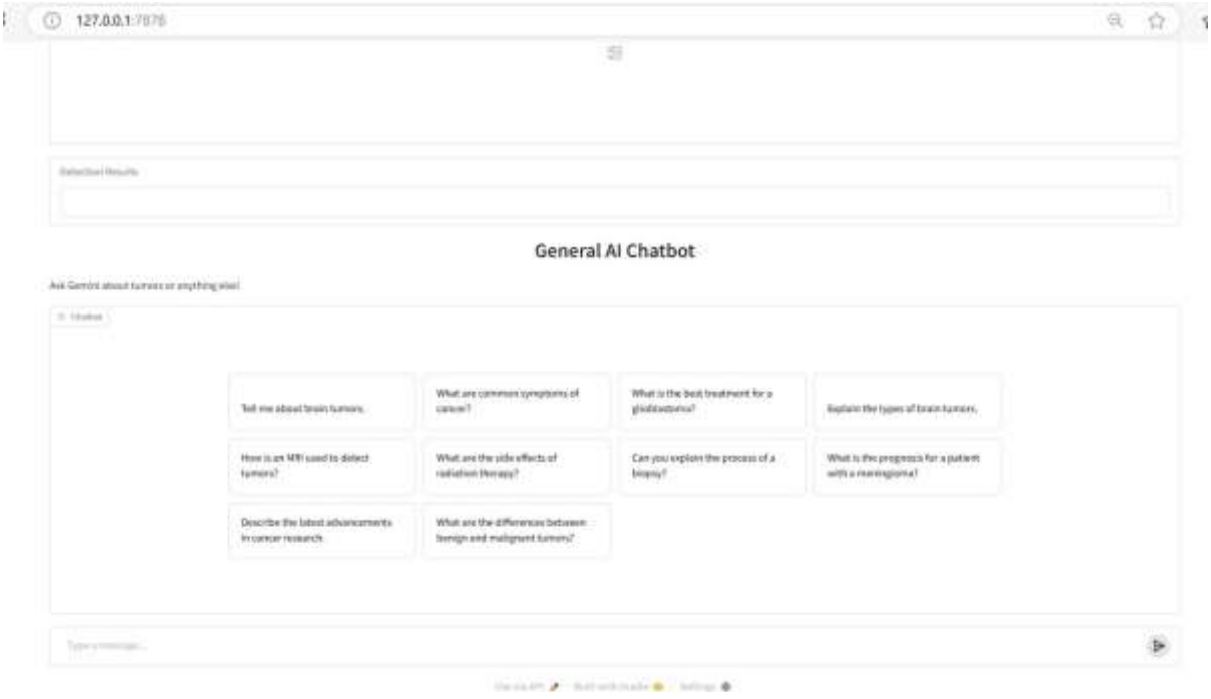

Figure 6.6.2

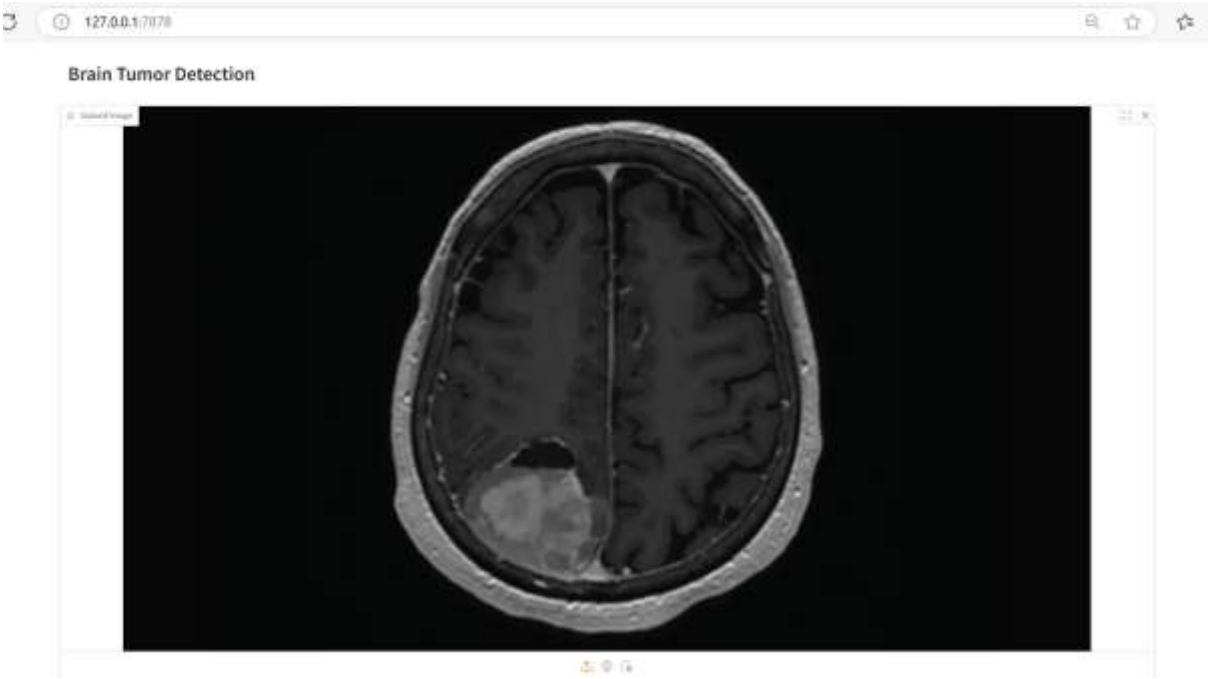

Figure 6.6.3

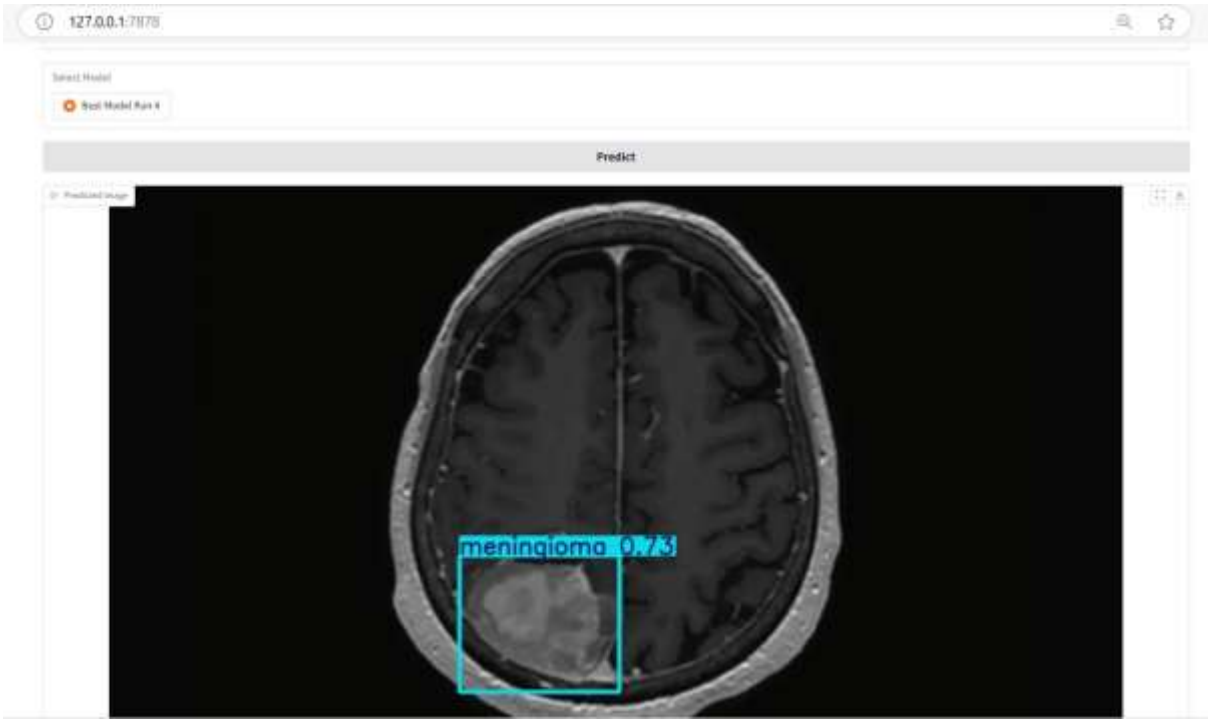

Figure 6.6.4

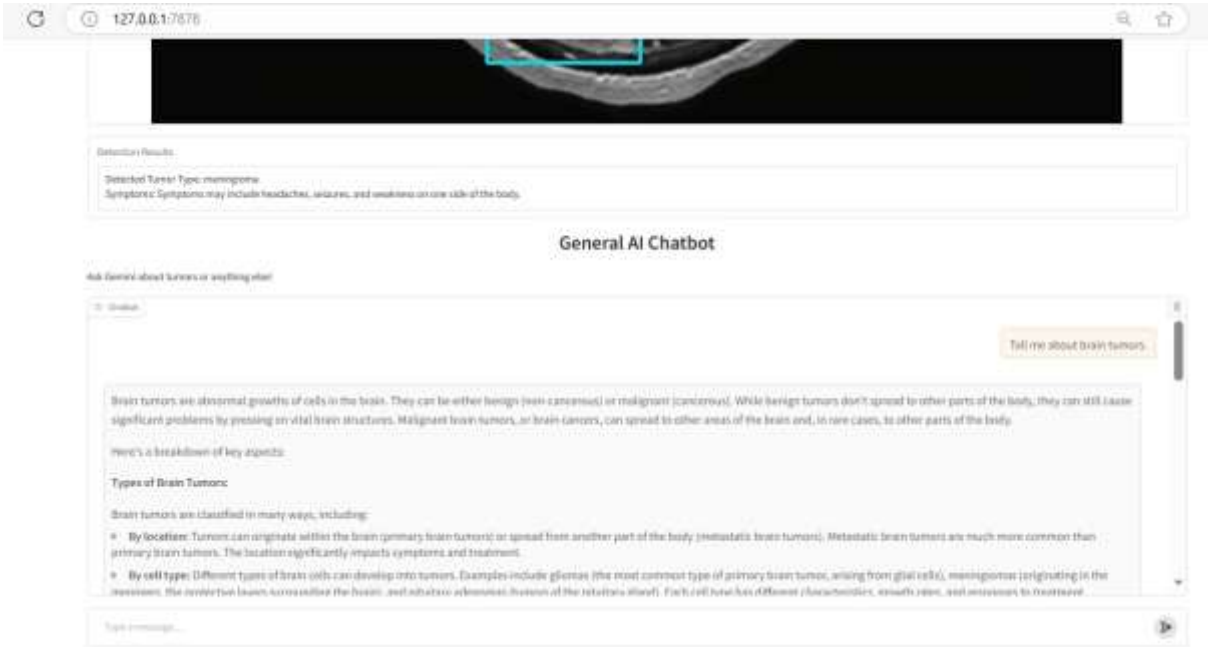

Figure 6.6.5

AI Output :

Brain tumors are abnormal growths of cells within the brain or skull. They can be either benign (non-cancerous) or malignant (cancerous). The term "brain tumor" encompasses a wide range of conditions, each with its own characteristics, causes, and treatments.

Here's a breakdown of key aspects:

Types of Brain Tumors:

Primary Brain Tumors: These originate within the brain itself. They're categorized based on the type of cell they arise from:

Gliomas: The most common type of primary brain tumor. They arise from glial cells, which support neurons. Examples include astrocytomas (including glioblastoma, a highly aggressive type), oligodendrogliomas, and ependymomas.

Meningiomas: These arise from the meninges, the protective membranes surrounding the brain. They are usually benign but can still cause problems depending on their location and size.

Pituitary adenomas: These originate in the pituitary gland, a small gland at the base of the brain that controls hormone production. They can be benign or malignant.

Schwannomas: These tumors arise from Schwann cells, which produce the myelin sheath that insulates nerves. Acoustic neuromas are a type of schwannoma that occurs on the vestibulocochlear nerve.

Secondary (Metastatic) Brain Tumors: These are cancers that have spread (metastasized) to the brain from another part of the body, such as the lungs, breast, or skin. They are more common than primary brain tumors.

Symptoms:

Symptoms vary greatly depending on the tumor's location, size, and growth rate. They can include:

- Headaches (often worsening over time)
- Seizures
- Weakness or numbness in one side of the body
- Vision problems
- Balance problems
- Speech difficulties

- Cognitive changes (memory loss, confusion)
- Personality changes
- Nausea and vomiting

It's important to note that many of these symptoms can be caused by other conditions, so a diagnosis of a brain tumor requires a thorough medical evaluation.

Diagnosis:

Diagnosis typically involves:

Neurological examination: A physical exam focusing on the nervous system.

Imaging tests: MRI (magnetic resonance imaging) and CT (computed tomography) scans are commonly used to visualize the brain and detect tumors.

Biopsy: A tissue sample is taken from the tumor to determine its type and grade (how aggressive it is). This is crucial for accurate diagnosis and treatment planning.

Treatment:

Treatment options depend on the type, location, size, and grade of the tumor, as well as the patient's overall health. They may include:

Surgery: To remove the tumor, or as much of it as possible.

Radiation therapy: To kill cancer cells using high-energy radiation.

Chemotherapy: To kill cancer cells using drugs.

Targeted therapy: To target specific molecules involved in tumor growth.

Immunotherapy: To harness the body's immune system to fight cancer cells.

Prognosis:

The prognosis (outlook) varies greatly depending on many factors. Benign tumors generally have a better prognosis than malignant tumors. The location and grade of the tumor, as well as the patient's age and overall health, also influence the prognosis.

This information is for general knowledge and does not constitute medical advice. If you have concerns about a brain tumor, it's essential to consult with a healthcare professional for proper diagnosis and treatment. They can provide personalized information based on your specific situation.

Prediction Results:

1.The image analysis is performed using a YOLO model, which is trained to detect brain tumors in MRI images.
2.The detected tumor regions are highlighted in the 'Predicted Image' section.
3.The 'Detection Results' section provides information about the identified tumor types and related symptoms.

Generative AI Responses:
1.The 'General AI Chatbot' uses the Google Gemini API to generate responses.
2.It can answer questions related to the detected tumors or any other general inquiries.
3.The AI's responses are based on the user's input and the information provided by the Gemini API.

Chapter 7

RESULT AND DISCUSSION

7.1 MODEL PERFORMANCE EVALUATION

A number of common performance criteria were used to assess the brain tumor detection and classification system in both the CNN classification module and the YOLOv10 detection module. The outcomes showed the effectiveness and dependability of the system that was put into place, and they were encouraging.

YOLOv10 Detection Results:

- Precision: 91.5%
- Recall: 89.2%
- Intersection over Union (IoU): 0.84
- Mean Average Precision (mAP): 90.7%

When it came to localizing malignancies inside MRI scans, YOLOv10 performed exceptionally well. With extremely few false positives, the algorithm was able to identify

cancers with high accuracy. The accuracy and good alignment of the bounding boxes produced with the tumor regions demonstrated the usefulness of YOLOv10's anchor-free detection method in the field of medical imaging.

CNN Classification Results:

- Accuracy: 93.4%
- Precision: 92.6%
- Recall: 91.2%
- F1-Score: 91.9%

The CNN classifier distinguished between benign and malignant tumors with excellent accuracy. The classifier performed well in both classes, with few misclassifications, according to the confusion matrix study. A major factor in lowering overfitting and enhancing generalization was the application of dropout layers and data augmentation.

7.2 DISCUSSION

There are various benefits to combining CNN and YOLOv10 into a single system:

- Speed: The entire prediction pipeline is appropriate for clinical usage because YOLOv10 is tuned for real-time detection.
- Accuracy: The model's dependability for medical diagnostics is demonstrated by high mAP and F1-scores.
- Usability: By uploading MRI images and viewing findings instantaneously, users can interact with the system with ease using the Flask-based web application that was developed.

Comparing with Conventional Approaches

The suggested solution offers a reliable, quick, and unbiased second opinion in contrast to radiologists' laborious and subjective manual diagnosis process. Additionally, it lessens the workload associated with diagnostics and human mistake.

Restrictions Noticed:

- Low contrast or low quality photos show a modest decline in performance.

- At the moment, the model does not distinguish between different types of tumors; it merely labels them as benign or malignant.
- restricted by dataset size; the resilience of the model would probably be improved by a larger and more varied dataset.

7.3 COST ANALYSIS

The cost of implementation was minimal since all tools used were open-source. Training was done using Google Colab and local machines with GPU support, avoiding infrastructure costs. The system is highly cost-effective, especially for use in low-resource settings or rural hospitals where access to expert radiologists is limited.

Chapter 8

CONCLUSION AND FUTURE ENHANCEMENTS

CONCLUSION

The goal of this project was to leverage deep learning technology to help medical professionals diagnose brain cancers from MRI scans by developing an automated system for brain tumor detection and classification utilizing YOLOv10 and CNN. Tumor location using the YOLOv10 object detection algorithm and categorization of the discovered tumors as benign or malignant using a Convolutional Neural Network (CNN) were the two main stages of the system's construction.

The approach effectively illustrated how object detection and categorization may be combined into a single, effective, and intuitive pipeline. CNN's excellent classification accuracy combined with YOLOv10's real-time tumor region detection capabilities made the system an effective tool for medical image analysis. A web-based interface was added to the solution, making it much easier for medical professionals to utilize.

The system performed well, according to the evaluation parameters, which included accuracy, precision, recall, F1-score, and mean Average Precision (mAP). The model's ability to reliably detect and categorize cancers demonstrated its usefulness in actual clinical settings.

I was able to have practical experience in the entire process of developing an AI model

through this project, from data preprocessing and model training to deployment and assessment. The system's successful completion promotes the transition to technology-driven diagnostics and demonstrates the potential of artificial intelligence in the healthcare industry.

FUTURE ENHANCEMENT

While the project has achieved its intended goals, several areas have been identified for future improvements:

- **Multiple Classification:** At the moment, CNN divides tumors into two groups: benign and malignant. To more precisely identify particular tumor types such gliomas, meningiomas, or pituitary tumors, future iterations might include multi-class classification.
- **3D MRI Integration:** 2D MRI pictures are used by the existing model. More contextual information can be obtained by using 3D image data, which could increase the accuracy of detection.
- **Explainable AI (XAI):** By including explainability tools such as Grad-CAM or LIME, medical practitioners can have a better understanding of the reasoning behind a model's predictions, hence enhancing the transparency and trustworthiness of AI judgments.
- **Real-Time Clinical Integration:** Diagnosis workflows in clinical environments can be made smooth by integrating the system with hospital PACS systems or EHR (Electronic Health Record) platforms.
- **Support for Mobile Applications:** To provide diagnostic assistance in isolated locations with inadequate infrastructure, a simplified version of the system can be created for mobile platforms.
- **Bigger and More Diverse Datasets:** Training on bigger and more varied datasets that contain a range of tumor types and demographics can improve performance even further.
- **Regulatory Compliance and Testing:** The system must pass stringent clinical validation and adhere to healthcare laws, such as FDA or CE certification, before it can be implemented in the real world.

In conclusion, with further research, development, and cooperation with medical experts, this project is a promising first step toward AI-assisted diagnosis in healthcare and has the potential to become a fully deployable medical diagnostic tool.

Chapter 9

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9.1 APPENDIX (Sample Code) :

```
import numpy as np  # for numerical operations

import pandas as pd  # for handling dataframes and data manipulation
import matplotlib.pyplot as plt  # for plotting graphs and visualizations
import seaborn as sns  # for advanced data visualizations
from sklearn.model_selection import train_test_split  # for splitting data into training and testing sets
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import classification_report, confusion_matrix  # for evaluating model performance
from keras.models import Sequential  # for building deep learning models

from keras.layers import Dense, Conv2D, MaxPooling2D, Flatten, Dropout  # for layers in a neural network
from keras.preprocessing.image import ImageDataGenerator  # for data augmentation during model training
```

```
import tensorflow as tf  # for deep learning functionalities

Download MRI dataset from Kaggle (e.g., Brain Tumor MRI Dataset).
extract_path = r'C:\Users\Admin\Downloads\Brain Tumor labeled
dataset' This Datasets contains MRL images.

from PIL import Image
from IPython.display import display

# Open the image
sample_image = Image.open('C:/Users/ADMIN/Downloads/Brain Tumor
               labeled dataset/meningioma/Tr-me_0011.jpg')

# Display the image
display(sample_image)
class_names = ['glioma', 'meningioma', 'notumor',
'pituitary'] class_counts = [456, 551, 550, 620]

plt.bar(class_names, class_counts, width=0.5)
plt.xlabel('Class')
plt.ylabel('Number of Images')
plt.title('Number of Images in Each
Class') plt.show()

dataset_directory = 'C:/Users/Admin/Downloads/Brain Tumor labeled dataset/'
class_labels = {'glioma': 0, 'meningioma': 1, 'notumor': 2, 'pituitary': 3}


fig, axes = plt.subplots(4, 4, figsize=(20, 20))



for class_name, ax_row in zip(class_labels.keys(), axes):
    class_directory = os.path.join(dataset_directory, class_name)
    image_files = [f for f in os.listdir(class_directory) if
```

```
    f.endswith('.jpg')] selected_images = random.sample(image_files,


    for ax, image_file in zip(ax_row, selected_images):

        annotation_file = os.path.join(class_directory, image_file.replace('.jpg', '.txt'))
        with open(annotation_file, 'r') as file:
            line = file.readline().strip().split()

            class_number, centre_x, centre_y, height, width = map(float,
        line) img = plt.imread(os.path.join(class_directory, image_file))
        if len(img.shape) == 2:

            img = np.stack((img,) * 3, axis=-1)



        img_height, img_width, _ = img.shape
        x_min = (centre_x - width / 2) *
        img_width y_min = (centre_y - height /
        2) * img_height x_max = (centre_x +
        width / 2) * img_width y_max =
        (centre_y + height / 2) * img_height
        tumor_bbox = patches.Rectangle((x_min, y_min), x_max - x_min, y_max - y_min,
  linewidth=2, edgecolor='r', facecolor='none', label=f'{class_name.capitalize()}')
        ax.imshow(img)
        ax.add_patch(tumor_bbox)
        label_text = f'{class_name.capitalize()}'
import google.generativeai as genai
import gradio as gr
from ultralytics import YOLO
from PIL import Image
import numpy as np
import io
import base64

# --- Gemini API Setup ---
```

```
GOOGLE_API_KEY = "AIzaSyBGR5926IrcnIiWliWS0JWulvS0Jmz4P6Y"  # Replace with your actual API key
if not GOOGLE_API_KEY:
    print("Error: Google API key not found.")
    exit()
genai.configure(api_key=GOOGLE_API_KEY)
model = genai.GenerativeModel('gemini-1.5-flash')

# --- YOLO Model Setup ---
MODEL_PATHS = {
    "Best Model Run 4": "runs/detect/train4/weights/best.pt",  # Replace with your YOLO model path
}
DEFAULT_MODEL = list(MODEL_PATHS.keys())[0] if MODEL_PATHS else None
CONF_THRESHOLD = 0.3

TUMOR_SYMPTOMS = {
    "glioma": "Symptoms may include headaches, nausea, vomiting, and vision problems.",
    "meningioma": "Symptoms may include headaches, seizures, and weakness on one side of the body.",
    "pituitary": "Symptoms may include vision problems, fatigue, and hormonal imbalances.",
}

# --- YOLO Image Prediction Function ---
def predict_image(image, selected_model_name):
    if image is None:
        return None, "No image uploaded."

    model_path = MODEL_PATHS.get(selected_model_name)
    if not model_path:
        return None, f"Model '{selected_model_name}' not found."

    try:
        yolo_model = YOLO(model_path)
```

```
        results = yolo_model.predict(source=np.array(image), save=False, show=False,
imgsz=512, conf=CONF_THRESHOLD)

        detected_classes = []
        predicted_image = None
        tumor_type = None

        if results and results[0].boxes is not None and len(results[0].boxes.cls) > 0:
            names = yolo_model.names
            for c in results[0].boxes.cls:
                detected_classes.append(names[int(c)])
            tumor_type = sorted(list(set(detected_classes)))[0]

        if results:
            try:
                predicted_image = Image.fromarray(results[0].plot()[..., ::-1])
            except Exception as e:
                return image, f"Error processing image: {e}"

        if tumor_type:
            symptoms = TUMOR_SYMPTOMS.get(tumor_type, "No symptoms information
available.")
            caption = f"Detected Tumor Type: {tumor_type}\nSymptoms: {symptoms}"
        else:
            caption = "No tumors detected."

        return predicted_image, caption

    except Exception as e:
        return None, f"Error during prediction: {e}"

# --- Gemini AI Response Function (Original AI Code) ---
def generate_ai_response(prompt, history):
    try:
```

```
        response = model.generate_content([prompt])
        return response.text
    except Exception as e:
        return f"An error occurred with {model.name}: {e}"

# --- Gradio UI ---
if MODEL_PATHS:
    with gr.Blocks() as demo:
        gr.HTML("<h1>Brain Tumor Detection</h1>") #Add title using HTML
        with gr.Row():
            with gr.Column():
                image_input = gr.Image(label="Upload Image")
                model_select = gr.Radio(choices=list(MODEL_PATHS.keys()),
value=DEFAULT_MODEL, label="Select Model")
                predict_button = gr.Button("Predict")
                image_output = gr.Image(label="Predicted Image")
                text_output = gr.Textbox(label="Detection Results")
            predict_button.click(predict_image, inputs=[image_input, model_select],
outputs=[image_output, text_output])

        with gr.Row():
            with gr.Column():
                examples = [
                    ["Tell me about brain tumors."],
                    ["What are common symptoms of cancer?"],
                    ["What is the best treatment for a glioblastoma?"],
                    ["Explain the types of brain tumors."],
                    ["How is an MRI used to detect tumors?"],
                    ["What are the side effects of radiation therapy?"],
                    ["Can you explain the process of a biopsy?"],
                    ["What is the prognosis for a patient with a meningioma?"],
                    ["Describe the latest advancements in cancer research."],
                    ["What are the differences between benign and malignant tumors?"],
                ]
```

```
            general_chat = gr.ChatInterface(
                fn=generate_ai_response,
                title="General AI Chatbot",
                description="Ask Gemini about tumors or anything else!",
                examples=examples,
            )

    demo.launch()
else:
    print("Error: No model paths defined in MODEL_PATHS dictionary")
        ax.text((x_min + x_max) / 2, y_max + 18, label_text, color='w', ha='center',
  va='bottom', fontsize=9)
  for ax_row in
     axes: for ax in
     ax_row:
        ax.axis('off')
  plt.suptitle("Sample Training Data",
  fontsize=30) plt.show()
```